
A FORMAL PROOF OF A CONSERVATION LAW FOR THE RIEMANN HYPOTHESIS

*The Functional Equation Is Provably Silent on the
Eigenspace Where the Hypothesis Lives, and the
Hypothesis Is Bracketed Between Two Theorem-Grade
Walls*

Preprint for Peer Review · Analytic Number Theory and Mathematical Physics

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ABSTRACT

We give a single consolidated account of what the analytic structure of the Riemann zeta function does and does not settle about the location of its non-trivial zeros, and we settle the relation between the two exactly. The work has four parts and we mark the grade of each. First, a geometric reformulation, stated at the strength it earns. The Berry-Keating operator $H = -i(x \, d/dx + 1/2)$, which on the additive space $L^2(\mathbb{R}^+, dx)$ carries a one-parameter family of self-adjoint extensions and therefore no canonical realization, is unitarily equivalent to the dilation generator $H' = -i \, x \, d/dx$ on the multiplicative Hilbert space $L^2(\mathbb{R}^{+*}, dx/x)$ carrying the Haar measure of the dilation group, where H' is uniquely self-adjoint by Stone's theorem with no boundary parameter. The deficiency-index ambiguity is dissolved by the change of geometry; the operator certifies a unique spectral arena, not the location of the zeros. Second, a localization. A regularized Cauchy-Stieltjes functional built from $-\zeta'/\zeta$ vanishes unconditionally in the safe regime to the right of the critical line; a Hardy-space equivalence reformulates the Riemann Hypothesis losslessly as a single causal-Fourier-support condition on one explicit tempered distribution; and the obstruction that blocks the safe vanishing from extending to the critical line is identified exactly as one object, the odd-symmetric Fourier component of the boundary distribution, equivalently the imaginary part of $-\zeta'/\zeta$ on the critical line, which the functional equation does not constrain and which three natural closure routes cannot supply without circularity. Third, the consolidating contribution: an identity and a two-sided barrier. The identity proves that determining the odd-part object is logically equivalent to the Riemann Hypothesis, so the obstruction is not adjacent to the problem but identical to it. The barrier brackets that object from two sides, each side a theorem. On the even side the entire machinery generated by the functional equation, being symmetric under the reflection of the critical line, is structurally incapable of determining an odd-parity object, so no refinement of the functional-equation toolkit can reach it; the even channel undershoots. On the odd side the natural positivity structure that does enter the odd channel, the de Branges Hilbert-space approach, imposes a condition strictly stronger than the Riemann Hypothesis that the zeta function fails to satisfy, established by Conrey and Li; the odd channel overshoots into falsity. The obstruction sits in the gap between an even-channel undershoot and an odd-channel overshoot, and the direction of each miss is named. The even-side miss is sharpened to a conservation law: the functional equation is an involution acting on the critical line as the parity reflection, the hypothesis lives in its odd eigenspace, and by the spectral theorem no consequence of the functional equation can carry information into that eigenspace, so the closure requirement takes the exact form of a second symmetry of the zeta function acting nontrivially on the conserved eigenspace, the structure realized in the proven function-field case by the action of Frobenius on cohomology. A magnitude bound of Lindelöf type, often listed as an

independent route, is shown to be parity-even and therefore to fall inside the even-side barrier rather than outside it. Fourth, a universality analysis identifying the equilibrium class of the prime field by four orthogonal critical exponents, with a measurement of which exponents discriminate the Riemann Hypothesis from its negation and which are truth-invariant across it, rendered as vanishing mutual information; and an independent structural witness from atomic physics, that the periodic table, a physical spectrum whose shell structure is fixed by the spatial rotation group and whose ordering runs over the integer counting of nuclear charge with its filling order an empirical effective-theory rule, never invokes the factorization of the integers into primes, confirming that the multiplicative prime channel in which the Riemann Hypothesis lives is a structurally distinct sector from the spatial-and-counting channel that suffices elsewhere. The consolidated picture is a measurement and a bracket. The distance from the unconditional Cauchy anchor to the critical line is one object wide; the object is the odd-part component; the component is the Riemann Hypothesis itself; and it is bracketed between two named, theorem-grade failure-directions, the even channel that cannot reach it and the odd channel that overshoots it. We make no claim of a proof. The boldest statement is the measurement: the problem is one object wide, the object is located, and the two standard channels miss it in opposite directions, the even channel falling short of it and the odd channel overshooting it. The bracket encloses these two standard channels exactly; a route that closes the hypothesis must either thread the gap between them, entering the odd channel without the even channel's parity limitation and without the natural odd construction's over-demand, or step outside the odd-side wall by building a modified construction the Conrey-Li counterexample does not reach. Either way the two named channels are walled off, and the work that remains is precisely characterized. We close by stating the determination in the two registers in which the hypothesis has a status. In the strict formal register of syntactic proof it is open, localized to the odd-part object, and bracketed between the two walls. In the physical-structural register it is the convergent determination of four mutually independent axes, the formal equivalence, the equilibrium universality class, three arithmetic-topological invariants of the integers, and the universal physics of order-parameter localization, which survive removal of every shared covariate but one. The single covariate that does not subtract out, the identification of the spectrum of the self-adjoint operator with the imaginary parts of the zeros, is identically the strict-register gap, so the two registers meet on one open object. The physical-structural register returns a convergent determination that points to the hypothesis as true, pending the determination of the odd-part component of $-\zeta'/\zeta$ on the critical line. That odd-part component is, pointwise, the regularization width of each pole of $-\zeta'/\zeta$ on the line and so the record of every deviation of a zero from it, which makes its determination logically the hypothesis; and the explicit formula being a conserved identity between the zeros and the primes excludes any prime-side identity that would determine it independently, since

such an identity is either insensitive to the deviations and true regardless of the hypothesis, or sensitive to them and reducible to the hypothesis through the conservation. The one determination the structure does not foreclose is one sourced from outside the prime-zero ledger, an operator exhibiting the zeros as its spectrum given independently of the zeta function, and none is known. The originating framework and its interpretive layer are confined to an appendix and are load-bearing for no result in the body.

Keywords: Riemann Hypothesis, Hilbert-Polya conjecture, multiplicative Hilbert space, Haar measure, Mellin transform, Cauchy-Stieltjes transform, Hardy space, Paley-Wiener-Schwartz theorem, even-odd parity decomposition, functional equation, de Branges spaces, Conrey-Li positivity, Selberg variance theorem, Gaussian Unitary Ensemble, universality class, mutual information.

1. INTRODUCTION

A real distinction holds between the part of the zeta function's structure that the functional equation controls and the part it does not. It is felt as the gap between a hundred and sixty-five years of overwhelming numerical confirmation and the continued absence of a proof, and at the level of structure that gap is sound and locatable. The same distinction is repeatedly mistaken for an argument that the proof is merely a matter of further effort with the standard toolkit, and the mistake is consequential: it leaves the field expecting that some refinement of functional-equation methods, some sharper bound or sharper symmetry, will eventually close the question. This paper consolidates, into one standalone account, what the structure establishes and what it cannot. It states the analytic facts at their correct strength, organizes the question of zero location into the layers that do and do not share a fate, and proves with precision that the distance from the safe, unconditional vanishing of a Cauchy functional to the critical line is exactly one object wide, that the object is identical to the Riemann Hypothesis itself, and that the object is bracketed between two named failure-directions: the even-symmetric machinery cannot reach it, and the natural odd-channel positivity overshoots it into a condition the zeta function refutes.

Riemann reached the edge of this bracket in 1859. He built the even-symmetric machinery himself, the functional equation and the symmetrization of the completed function, and from it he saw the zeros on the line and computed them by hand far past what he published, as the later recovery of his working formulas confirms. He then stated the hypothesis without proof. The structural reading offered here is that the persistence of the problem since 1859 is not a record of insufficient cleverness but the signature of an obstruction that the even-symmetric construction can approach and cannot cross, because the answer is parity-orthogonal to the symmetry Riemann discovered. He stood at the even-side wall and saw across it. This paper names that wall, names the wall on the far side, and measures the gap between them.

The account is built so that each part carries the next. A geometric reformulation fixes the spectral arena in which the question is correctly posed. A Cauchy anchor supplies the unconditional fact, true with no hypothesis on zero locations. A Hardy-space equivalence translates the Riemann Hypothesis losslessly into a single causal-support condition on one distribution. And the consolidating work is an identity and a two-sided barrier: an identity locating the single obstruction and proving it identical to the Riemann Hypothesis, and a barrier proving the obstruction unreachable from either standard direction, the even channel that undershoots and the odd channel that overshoots into falsity. The layers and the obstruction are the same structure seen from two sides. The determination says where each layer lands; the barrier says why the open layer cannot be moved to a proof from either channel that settles the closed layers.

The parts carry different grades, and we state each. The reformulation of Section 3 is a unitary-equivalence calculation whose core is Stone's theorem; it dissolves a domain ambiguity and certifies a unique arena, nothing more. The Cauchy anchor of Section 4 is a contour argument, unconditional in the safe regime, attributed to standard analytic number theory. The Hardy-space equivalence of Section 4 is a Paley-Wiener-Schwartz computation establishing a lossless reformulation. The identity of Section 6 and the two-sided barrier of Section 7 are the consolidating contribution and are stated at theorem grade with their scopes fenced exactly; the even-side wall is proved here, and the odd-side wall is the published theorem of Conrey and Li. The universality analysis of Section 8 carries the grade its evidence supports, heuristic and inductive, with a precise measurement of which of its components discriminate the hypothesis and which do not.

The boldest true statement this paper makes is not that the Riemann Hypothesis is proven. It is that the hypothesis has been measured: its distance from an unconditional analytic fact is one object wide, that object is the hypothesis itself, and the object is bracketed between two theorem-grade walls, the even channel that provably falls short of it and the odd channel that provably overshoots it into a falsehood. This is a stronger and more durable claim than a contested proof, because nothing in it can be overturned by a sharper estimate from either channel: the gap it names is the hypothesis, and the two walls that bracket it are theorems. The bracket encloses the two standard channels exactly. A route that closes the hypothesis must either thread the gap between the two walls or step outside the odd-side wall by a modified construction the published counterexample does not reach, and Section 7 characterizes both alternatives precisely; neither is the refinement of a standard channel that the field has expected. No result in the body depends on any metaphysical commitment. The originating framework and its interpretation are confined to Appendix B.

2. WHAT THE PAPER ASSUMES AND WHAT IT PROVES

We fix the boundary at the outset so that no reader mistakes the register of any claim. The paper assumes the standard theory of the Riemann zeta function: the Dirichlet series and Euler product in the half-plane of absolute convergence, the meromorphic continuation, the functional equation for the completed function ξ , and the standard zero-free region of Hadamard and de la Vallée Poussin. It assumes elementary operator theory: Stone's theorem, the Mellin and Fourier transforms, and the Paley-Wiener-Schwartz characterization of Hardy-space boundary values. It assumes the de Branges theory of Hilbert spaces of entire functions and the result of Conrey and Li (2000) on the failure of the de Branges positivity conditions for the zeta function. It assumes the established results of equilibrium critical phenomena cited in Section 8. It assumes nothing further. In particular it assumes no proposition internal to the originating framework, and it assumes neither the Riemann Hypothesis nor its negation.

The paper proves the following. That the question of zero location separates into components with distinct fates, of which the spectral arena and the unconditional Cauchy anchor are settled and the location itself is open. That the obstruction blocking the anchor from reaching the critical line is a single object, the odd-part Fourier component of the boundary distribution. That this obstruction, closed, is logically equivalent to the Riemann Hypothesis. That the obstruction is bracketed by two theorem-grade walls: it cannot be determined by the functional equation or any extension of it, and it is not supplied by the natural odd-channel positivity, the de Branges approach, which imposes a condition strictly stronger than the Riemann Hypothesis that the zeta function fails to satisfy by the theorem of Conrey and Li. And that the even-side wall is not merely a barrier against a technique family but a conservation law: the functional equation is an involution, its action on the critical line is the parity reflection, the Riemann Hypothesis lives entirely in the -1 eigenspace of that involution, and by the spectral theorem the symmetry acts trivially on its own $+1$ eigenspace and therefore carries, provably, no information into the eigenspace where the hypothesis lives. From this the closure requirement takes its sharpest form, stated as a necessary condition: closure requires a second structural symmetry of the zeta function, independent of the functional equation, acting nontrivially on the conserved eigenspace, which is exactly the structure that closes the proven function-field analogue through the action of Frobenius on cohomology. Around these it states, at marked and limited strength, a geometric reformulation, a universality analysis, and an independent structural witness from atomic physics. The reformulation is a modeling and clarifying instrument; the identity, the two-sided barrier, and the conservation law are its load-bearing yield. A reader who sets aside both the operator and the physics retains the identity, the barrier, and the conservation theorem, since each is stated in

standard analytic terms and none depends on the operator's machinery or the universality picture.

3. THE MULTIPLICATIVE GEOMETRY AND THE SPECTRAL ARENA

We present the operator reformulation, fix exactly what it certifies, and state at the point of definition that it certifies a unique arena and not the location of the zeros.

3.1 Construction

The multiplicative group $(\mathbb{R}^{+*}, \times)$ is locally compact abelian, with unique Haar measure dx/x up to scaling. Define the Hilbert space

$$H_{\text{mult}} := L^2(\mathbb{R}^{+*}, dx/x) = \{ f : \mathbb{R}^+ \rightarrow \mathbb{C} \text{ measurable} : \int_0^\infty |f(x)|^2 dx/x < \infty \},$$

with inner product $\langle f, g \rangle = \int_0^\infty f(x) \overline{g(x)} dx/x$. The dilation operator $U_\lambda : f(x) \mapsto f(\lambda x)$, for each $\lambda > 0$, is unitary on H_{mult} , since the substitution $y = \lambda x$ leaves dx/x invariant:

$$\langle U_\lambda f, U_\lambda g \rangle = \int_0^\infty f(\lambda x) \overline{g(\lambda x)} dx/x = \int_0^\infty f(y) \overline{g(y)} dy/y = \langle f, g \rangle.$$

The Haar invariance of dx/x is the structural reason for the unitarity. No corresponding unitarity holds on $L^2(\mathbb{R}^+, dx)$, where $\|U_\lambda f\|^2 = \lambda^{-1} \|f\|^2$. The additive Lebesgue space is geometrically unsuited to dilations; the multiplicative Haar space is geometrically natural for them, and it is the Hilbert space whose inner product is invariant under the multiplicative symmetry group of the integers.

3.2 Stone's Theorem and Unique Self-Adjointness

The one-parameter family $\{ U_t : f(x) \mapsto f(e^t x), t \in \mathbb{R} \}$ is a strongly continuous unitary representation of $\mathbb{R}$ on H_{mult} . By Stone's theorem (Reed and Simon, 1975, Theorem VIII.7), it possesses a unique self-adjoint generator $H' := -i x d/dx$, with domain $D(H') = \{ f \in H_{\text{mult}} : x f(x) \in H_{\text{mult}} \text{ in the distributional sense} \}$. Self-adjointness follows from the unitarity of the underlying group representation; no boundary condition is imposed externally, and the deficiency-index ambiguity of the additive realization is absent.

Theorem 3.1 (Mellin unitarity). Define the Mellin transform on H_{mult} by $(\mathcal{M} f)(\tau) = \int_0^\infty f(x) x^{-i\tau} dx/x$. Then $\mathcal{M}$ extends to a unitary isomorphism $\mathcal{M} : H_{\text{mult}} \rightarrow L^2(\mathbb{R}, d\tau/2\pi)$, and under $\mathcal{M}$ the operator H' is diagonalized as multiplication by the real variable τ : $(\mathcal{M} H' \mathcal{M}^{-1} \varphi)(\tau) = \tau \varphi(\tau)$.

Proof. The substitution $u = \log x$ converts H_{mult} to $L^2(\mathbb{R}, du)$ with operator $-i d/du$. The Plancherel theorem for the standard Fourier transform on $\mathbb{R}$ gives the unitarity. ■

Corollary 3.2. The spectrum $\sigma(H') = \mathbb{R}$, purely continuous and Lebesgue absolutely continuous; every spectral value is real.

Theorem 3.3 (Unitary equivalence to Berry-Keating). The map $J : L^2(\mathbb{R}^+, dx) \rightarrow H_{\text{mult}}$, $(J f)(x) = x^{\{1/2\}} f(x)$, is a unitary isomorphism, and under J the Berry-Keating operator $H = -i(x d/dx + 1/2)$ on $L^2(\mathbb{R}^+, dx)$ is conjugated to H' on H_{mult} : $J H J^{-1} = H'$.

Proof. Direct computation gives $(J^{-1} H' J f)(x) = x^{\{-1/2\}}(-i x) d/dx[x^{\{1/2\}} f(x)] = -i(x f'(x) + (1/2) f(x)) = (H f)(x)$. Unitarity of J follows from $\int_0^\infty |x^{\{1/2\}} f(x)|^2 dx/x = \int_0^\infty |f(x)|^2 dx = \|f\|_{L^2(\mathbb{R}^+, dx)}^2$. The (1,1) deficiency-index problem of the additive realization is absent on the multiplicative geometry because the Haar invariance fixes the domain uniquely. ■

3.3 What the Reformulation Certifies

We state precisely what Theorems 3.1 through 3.3 establish, since an inflated reading would propagate through every claim below. The reformulation certifies that the spectral problem is correctly posed on a unique arena: the candidate Hilbert-Polya operator is H' , it is uniquely self-adjoint with no boundary parameter, its spectrum is the real line, and it is diagonalized by the Mellin transform. This dissolves the extension ambiguity that forced prior spectral approaches to introduce external apparatus, absorbing walls, supersymmetric completions, or Planck-scale cutoffs, to select a realization. It does not, by itself, place a single zero on the critical line. The arithmetic content, the connection to the zeros of ζ , enters in the next section through the Cauchy construction, and the obstruction to closing the question survives the reformulation intact. The operator fixes where the question lives; it does not answer it. This is the analogue, in the analytic register, of a measuring instrument that is correctly calibrated and correctly placed but whose reading is still the open quantity.

4. THE CAUCHY ANCHOR AND THE HARDY-SPACE EQUIVALENCE

This section supplies the unconditional fact and the lossless reformulation. The anchor is true with no hypothesis on zero locations; the equivalence translates the Riemann Hypothesis into a single causal-support condition.

4.1 The Cauchy Functional and Its Unconditional Vanishing

Define the regularized arithmetic distribution $\Omega_\varepsilon(\tau) = -(\zeta'/\zeta)(1/2 + \varepsilon + i\tau)$ for $\varepsilon > 0$, $\tau \in \mathbb{R}$, and its Cauchy-Stieltjes transform $G_\varepsilon(z) = (1/2\pi) \int_{-\infty}^{\infty} \Omega_\varepsilon(\tau)/(\tau - z) d\tau$ for $\text{Im}(z) > 0$. The transform is well-defined for $\text{Im}(z) > 0$ because in the regime $\varepsilon > 1/2$ the integrand has no singularity in the closed lower half τ -plane. For τ in that half-plane, written $\tau = \tau_R - i y$ with $y \geq 0$, the argument $s = 1/2 + \varepsilon + i\tau$ has $\text{Re}(s) = 1/2 + \varepsilon + y$, which is at least $1/2 + \varepsilon > 1$; the line and the half-plane below it therefore sit strictly to the right of the critical strip, where $\zeta(s) = \prod_p (1 - p^{-s})^{-1} \neq 0$ by Euler-product non-vanishing, so $-\zeta'/\zeta$ is holomorphic there with no poles. We do not invoke the zero-free region of the strip for this: the regime $\varepsilon > 1/2$ places the entire closed lower half τ -plane outside the strip, where non-vanishing is the elementary Euler-product fact, and the anchor is deliberately stated at this modest, fully unconditional end. The work of closure is the inward extension across the strip, taken up in Section 4.4.

Theorem 4.1 (Anchor vanishing). For every $\varepsilon > 1/2$ and every $z \in \mathbb{C}$ with $\text{Im}(z) > 0$, $G_\varepsilon(z) = 0$.

Proof. Fix $\varepsilon > 1/2$ and z with $\text{Im}(z) > 0$. For τ in the closed lower half-plane, $\tau = \tau_R - i y$ with $y \geq 0$, the argument $s = 1/2 + \varepsilon + i\tau$ has $\text{Re}(s) = 1/2 + \varepsilon + y > 1$, placing s in the half-plane of absolute convergence, where $\zeta(s) = \prod_p (1 - p^{-s})^{-1} \neq 0$ by Euler-product non-vanishing. Hence $\Omega_\varepsilon(\tau) = -(\zeta'/\zeta)(s)$ is holomorphic throughout the closed lower half τ -plane and, by the absolutely convergent Dirichlet series $-\sum_n \Lambda(n) n^{-(1/2+\varepsilon)} n^{-i\tau}$ on the line and its bound $|\Omega_\varepsilon(\tau_R - iy)| \leq \sum_n \Lambda(n) n^{-(1/2+\varepsilon+y)} \leq \sum_n \Lambda(n) n^{-(1/2+\varepsilon)} < \infty$, is bounded on that half-plane. It is therefore the boundary value of a bounded holomorphic function on the lower half τ -plane, that is, an element of H^∞ of the lower half-plane. The conclusion is the projection property of the Hardy decomposition: the Cauchy-Stieltjes transform from above, $z \mapsto (1/2\pi) \int_{-\infty}^{\infty} \Omega_\varepsilon(\tau)/(\tau - z) d\tau$ with $\text{Im}(z) > 0$, annihilates the boundary values of functions holomorphic and bounded in the lower half-plane, since for such a function the kernel $1/(\tau - z)$ is, for each fixed z in the upper half-plane, the boundary value of a function holomorphic in the lower half-plane and decaying at infinity, with its only pole at $\tau = z$ lying in the upper half-plane and outside the lower-half domain of Ω_ε , so the pairing of a lower-half-plane Hardy class against the complementary projection is zero. Hence $G_\varepsilon(z) = 0$ for every z with $\text{Im}(z) > 0$. ■

Theorem 4.1 establishes a fixed analytic anchor: for every $\varepsilon > 1/2$ the Cauchy transform of $-\zeta'/\zeta$ along the vertical line at distance ε to the right of the critical line vanishes identically in the upper half-plane. The vanishing is unconditional; it assumes nothing about zero locations because in the safe regime no zeros lie in the integration domain at all. The strategy of closure is to extend the anchor from $\varepsilon > 1/2$ down to the limit $\varepsilon \rightarrow 0^+$. If the extension succeeds, the resulting identity $G_0(z) = 0$ for $\text{Im}(z) > 0$ is exactly the Hardy-space membership condition of Theorem 4.2, equivalent to the Riemann Hypothesis. Section 4.4 examines whether the extension is achievable and identifies the precise obstruction.

4.2 The Hardy-Space Equivalence

Theorem 4.2 (Hardy-space equivalence). The following are equivalent.

- (i) The Riemann Hypothesis: every non-trivial zero ρ of ζ satisfies $\text{Re}(\rho) = 1/2$.
- (ii) The function $-(\zeta'/\zeta)(1/2 + i\tau)$, regarded as a tempered distribution on $\mathbb{R}$, is the boundary value of a function in the Hardy space H^2 (lower half τ -plane).
- (iii) The distributional Fourier transform $F[-(\zeta'/\zeta)(1/2 + i\tau)](u)$ is supported on the causal half-line $[0, +\infty)$; equivalently its negative- u part vanishes. Under the hypothesis the support is exactly the prime-power set $\{k \log p : p \text{ prime}, k \geq 1\}$, whose least element is $\log 2$.

Proof. (i) $\implies$ (ii). Under the Riemann Hypothesis all non-trivial zeros lie on the critical line, so $-\zeta'/\zeta$ is analytic for $\text{Re}(s) > 1/2$ except for simple poles on $\text{Re}(s) = 1/2$ and the pole at $s = 1$; in the τ -coordinate $s = 1/2 + i\tau$ the function is analytic in $\text{Im}(\tau) < 0$. The Hardy norm is finite by the Prime Number Theorem and the standard bounds (Titchmarsh, 1986, §14), with Sobolev-type estimates extending the bound to the boundary in the distributional sense.

(ii) $\implies$ (i). A function in H^2 (lower half τ -plane) has no poles in the upper half-plane. Poles of $-(\zeta'/\zeta)(1/2 + i\tau)$ in the upper τ -half-plane correspond, via $s = 1/2 + i\tau$, to zeros $\rho = \beta + i\gamma$ with $\text{Im}(\tau_\rho) = 1/2 - \beta > 0$, that is $\beta < 1/2$. Their absence asserts no zero with $\beta < 1/2$; the functional equation maps zeros at β to zeros at $1 - \beta$, so absence at $\beta < 1/2$ forces absence at $\beta > 1/2$, and all non-trivial zeros satisfy $\beta = 1/2$.

(ii) $\iff$ (iii). This is the Paley-Wiener-Schwartz theorem for tempered distributions: a distribution on $\mathbb{R}$ is the boundary value of an H^2 (lower) function if and only if its Fourier transform is supported on the causal half-line. The threshold $\log 2$ arises because the von Mangoldt expansion of $-\zeta'/\zeta$ has its first non-zero coefficient at $n = 2$. ■

Theorem 4.2 is genuine content. It is a lossless translation of the Riemann Hypothesis from a statement about zero locations in a two-dimensional domain into a statement about the causal Fourier support of one explicit distribution on

the line, with the threshold fixed by the smallest prime. The three formulations are mutually convertible. The reformulation does not constitute a proof; it establishes that the Riemann Hypothesis is equivalent to a single causal-support condition on a single tempered distribution. The remaining task is to verify that condition, and Section 4.4 identifies why each natural route is circular.

4.3 The Even-Odd Decomposition

Decompose the Fourier transform $F = F[-(\zeta'/\zeta)(1/2 + i\tau)]$ into even and odd parts under the reflection $u \leftrightarrow -u$:

$$F(u) = F_{\text{even}}(u) + F_{\text{odd}}(u), \text{ with } F_{\text{even}}(-u) = F_{\text{even}}(u) \text{ and } F_{\text{odd}}(-u) = -F_{\text{odd}}(u).$$

Causal support of F , that is $F(u) = 0$ for $u < 0$, is equivalent to the relation

$$F_{\text{odd}}(u) = -F_{\text{even}}(u) \text{ for all } u < 0,$$

a Hilbert-transform-type relation in which the even part determines the odd part if and only if the function is in $H^2(\text{lower half-plane})$. The functional equation $\xi(s) = \xi(1 - s)$ imposes on $-(\zeta'/\zeta)(1/2 + i\tau)$ a symmetry under $\tau \leftrightarrow -\tau$ that fixes the even part F_{even} , via the digamma representation of the Gamma factor in ξ :

$$\text{Re}[-(\zeta'/\zeta)(1/2 + i\tau)] = (1/2) \text{Re}[\psi(1/4 + i\tau/2)] + (\log \pi)/2 + \text{polar contribution},$$

whose Fourier transform is computable in closed form (Appendix C) and yields the explicit even measure

$$F_{\text{even}}(u) \propto -(2\pi) e^{-|u|/2} / (1 - e^{-2|u|}) + \text{delta-function terms at } u = 0,$$

of Bose-Einstein type. The factor $(1 - e^{-2|u|})^{-1}$ is singular as $u \rightarrow 0$, and the displayed expression is understood in the principal-value (Hadamard finite-part) sense at $u = 0$, which is the regularization that renders it a tempered distribution; the delta-function terms at $u = 0$ carry the local part. No theorem below depends on this prescription, since Theorem 7.1 uses only the parity identity $|F_{\text{even}}(u)| = |F_{\text{even}}(-u)|$, which holds in the $|u|$ variable independently of the $u = 0$ behavior. This even-part transform does not have causal support: $|F_{\text{even}}(u)| = |F_{\text{even}}(-u)|$ at every $u \neq 0$. The functional equation alone is therefore not sufficient to establish the causal support of the total transform F . Everything the functional equation determines lives in the even part; the causal-support condition lives in the relation between even and odd; and the odd part is left free.

4.4 The Localized Obstruction

Localized Obstruction. The odd part $F_{\text{odd}}(u)$ of the Fourier transform of $-(\zeta'/\zeta)(1/2 + i\tau)$, equivalently $\text{Im}[-(\zeta'/\zeta)(1/2 + i\tau)]$ regarded as a real-valued tempered distribution on $\mathbb{R}$, is not constrained by the functional equation $\xi(s) =$

$\xi(1 - s)$ and is not derivable from the even part F_{even} alone. An independent analytic identity controlling F_{odd} is required to establish the Hardy-space equivalence and hence the Riemann Hypothesis.

The three natural closure routes are each circular without additional input.

Route A: naive Paley-Wiener extension. The Dirichlet series $\Omega_{\varepsilon}(\tau) = -\sum_n \Lambda(n) n^{-(1/2 + \varepsilon + i\tau)}$ has formal Fourier transform $\sum_n \Lambda(n) n^{-(1/2 + \varepsilon)} \delta(u - \log n)$, supported on $\{\log p^k\} \subset [\log 2, +\infty)$. For $0 < \varepsilon < 1/2$ the series no longer converges on the line, and the meromorphic continuation acquires poles at τ_{ρ} for each zero ρ with $\beta_{\rho} > 1/2 + \varepsilon$; whenever $\beta_{\rho} > 1/2 + \varepsilon$ the corresponding pole lies in the lower half τ -plane and contributes a residue term proportional to $e^{i\gamma_{\rho}u} \cdot e^{(\beta_{\rho} - 1/2 - \varepsilon)u}$, non-decaying for $u < 0$. The support claim therefore presumes the absence of zeros with $\beta_{\rho} > 1/2 + \varepsilon$, which, taking ε arbitrarily small, is the Riemann Hypothesis. The route assumes what is to be proved.

Route B: ε -constancy of G_{ε} . To show $\partial G_{\varepsilon} / \partial \varepsilon \equiv 0$ and conclude $G_0 = G_{\varepsilon} = 0$ by continuity, one needs the derivative, itself a Cauchy-Stieltjes transform of a Dirichlet-class function, to satisfy the same support claim; the circularity of Route A applies identically.

Route C: Sobolev weak-star limit. The convergence $\Omega_{\varepsilon} \rightarrow \Omega_0$ in a weighted Sobolev dual and the continuity of the Cauchy transform can both be made rigorous, but preservation of causal support through the limit requires that no pole migrate from the upper to the lower half τ -plane as ε decreases. Such migration occurs if and only if zeros exist with $\beta_{\rho} > 1/2$. The continuity argument transmits causal support only under the assumption that the support is already causal in the limit, which is the Riemann Hypothesis.

All three routes fail by one mechanism: in each case the missing constraint is control on the odd-symmetric component F_{odd} , independent of the functional equation and the carrier of the zero-location information. The candidate independent constraints in the literature each represent substantial open problems. Lindelöf-type bounds on $|\zeta(1/2 + it)|$ constrain the magnitude of $-\zeta'/\zeta$ but not its imaginary part. Rankin-Selberg automorphic moments give averaged estimates, not pointwise odd-part control. The de Branges positivity conditions supply a class of formal constraints whose verification has itself eluded proof. None has been closed. The status of Sections 3 through 4 is therefore exact: the spectral arena is fixed, the Cauchy anchor is unconditional in the safe regime, the Riemann Hypothesis is reformulated as a lossless causal-support condition, and the obstruction to closing it is a single object, the odd-part Fourier component, controlled by no standard symmetry of ζ . Sections 6 and 7 prove what this object is and why it is unreachable by the methods that fix the even part.

5. THE FORM-ANALYSIS DISTINCTION

We distinguish a symmetry of the zeta function considered as a property of the completed function from the analytic content that symmetry transmits to the boundary distribution on the critical line. The distinction is ordinary. The functional equation is a single exact relation, $\xi(s) = \xi(1 - s)$; the boundary distribution $-(\zeta'/\zeta)(1/2 + i\tau)$ is a tempered object with even and odd parts, and the question is how much of that object the relation determines. We record the distinction because the equivalence of Section 4, the characterization of Section 6, and the barrier of Section 7 all turn on the difference between what the functional equation is, an even symmetry under the critical-line reflection, and what the Riemann Hypothesis needs, control of an odd-parity component. An interpretive reading of the distinction appears in Appendix B and is optional.

6. THE OBSTRUCTION IS THE HYPOTHESIS

We identify, exactly, the object whose determination would close the question, and we prove that determining it is the Riemann Hypothesis. This is the hinge on which the consolidated determination of Section 10 turns.

The Riemann Hypothesis is the proposition that every non-trivial zero has real part one half. Theorem 4.2 reformulates it as the causal-support condition on F , and Section 4.3 reduces causal support to the single relation $F_{\text{odd}}(u) = -F_{\text{even}}(u)$ for $u < 0$, with F_{even} fixed by the functional equation. The remaining content is the determination of F_{odd} , the odd part of the boundary distribution.

Definition 6.1 (the odd-part closure). The odd-part closure, denoted Θ , is the proposition that the odd-symmetric Fourier component F_{odd} of $-(\zeta'/\zeta)(1/2 + i\tau)$ satisfies $F_{\text{odd}}(u) = -F_{\text{even}}(u)$ for all $u < 0$, equivalently that $\text{Im}[-(\zeta'/\zeta)(1/2 + i\tau)]$ is the boundary value, on the odd channel, of the Hardy-space relation that makes the total transform causal.

The closure Θ is precisely what must be added to the functional-equation determination of F_{even} to obtain the causal-support condition, and hence the Riemann Hypothesis. The functional equation supplies F_{even} ; Θ supplies the relation that ties F_{odd} to F_{even} on the negative half-line; conjoined, they yield causal support of F , which by Theorem 4.2 is the Riemann Hypothesis. Without Θ , the even-part determination is silent on F_{odd} , by Section 4.3, and the causal-support condition is undecided. The closure is therefore the entire logical distance between the unconditional even-part fact and the hypothesis.

Theorem 6.2 (the obstruction is the hypothesis). The odd-part closure Θ is logically equivalent to the Riemann Hypothesis.

Proof. ($\Theta \implies \text{RH}$.) Suppose $F_{\text{odd}}(u) = -F_{\text{even}}(u)$ for all $u < 0$. Then $F(u) = F_{\text{even}}(u) + F_{\text{odd}}(u) = F_{\text{even}}(u) - F_{\text{even}}(u) = 0$ for all $u < 0$, so F has causal support on $[0, +\infty)$. By Theorem 4.2(iii) $\implies$ (i), the Riemann Hypothesis holds, and the support is then the prime-power set $\{k \log p\}$, with least element $\log 2$.

($\text{RH} \implies \Theta$.) Suppose the Riemann Hypothesis. By Theorem 4.2(i) $\implies$ (iii), F has causal support, so $F(u) = 0$ for all $u < 0$, that is $F_{\text{even}}(u) + F_{\text{odd}}(u) = 0$ for $u < 0$, which is exactly $F_{\text{odd}}(u) = -F_{\text{even}}(u)$ for $u < 0$, the closure Θ . ■

The consequence is exact. The distance between the unconditional even-part determination supplied by the functional equation and the Riemann Hypothesis is one object wide, and that object is the odd-part closure Θ , which is identically the hypothesis. We are explicit about the depth of this identity and we do not inflate it. Unlike the Cook-Levin route that makes a physical no-shortcut premise identical to a complexity-class separation through the non-trivial machinery of NP-completeness, the equivalence $\Theta \iff \text{RH}$ is near-immediate once the Hardy-

space reformulation of Theorem 4.2 is in hand: it is the parity decomposition of a causal-support condition, read in both directions. Its role is not to be deep but to be exact. It fixes the target with no slack: any argument that proposes to cross from the functional-equation fact to the hypothesis must supply Θ , and supplying Θ is supplying the odd-part determination, which is the hypothesis. There is no shorter route and no route around, for the gap is exactly the closure and nothing less. The weight of the consolidating contribution is therefore carried not by this identity but by the two-sided bracket of the next section, which proves that Θ is enclosed between the even channel that cannot reach it and the odd-channel positivity that overshoots it into falsity. An argument that crosses the distance while claiming to use only the functional equation has committed the parity error of the even-side wall, and an argument that crosses it by the natural odd-channel positivity has run into the falsity of the odd-side wall; Section 7 makes both walls precise.

7. THE OBSTRUCTION IS BRACKETED BY TWO WALLS

The closure Θ is the hypothesis, by Theorem 6.2. We now prove that Θ is bracketed between two theorem-grade walls, one on each side of the channel it lives in. The even-side wall proves that the functional equation and every extension of it undershoot Θ : they cannot enter the odd channel at all. The odd-side wall proves that the natural construction which does enter the odd channel, the de Branges positivity structure, overshoots Θ into a condition the zeta function refutes. The obstruction sits in the gap between the two, and the direction of each miss is named. We give the even-side wall first, then the odd-side wall, then the bracket they jointly form.

7.1 The Even-Side Wall: the Functional Equation Undershoots

The even-side wall rests on a parity fact: the functional equation is even under the critical-line reflection, the closure is odd, and an even symmetry transmits no information to an odd-parity channel.

Theorem 7.1 (even-side wall, parity barrier). Let $\mathcal{F}$ be any analytic identity or family of identities derivable from the functional equation $\xi(s) = \xi(1 - s)$ together with the standard archimedean data (the Gamma factor, the digamma representation, and the polar terms at $s = 0, 1$). Every such $\mathcal{F}$, transported to the boundary distribution on the critical line via $s = 1/2 + i\tau$, is invariant under the reflection $\tau \leftrightarrow -\tau$, and therefore determines only the even-symmetric component F_{even} of the Fourier transform of $-(\zeta'/\zeta)(1/2 + i\tau)$. No such $\mathcal{F}$ determines the odd-symmetric component F_{odd} , and hence no such $\mathcal{F}$ supplies the closure Θ .

Proof. The functional equation is the single relation $\xi(s) = \xi(1 - s)$. Under the substitution $s = 1/2 + i\tau$, the reflection $s \leftrightarrow 1 - s$ becomes $\tau \leftrightarrow -\tau$, since $1 - (1/2 + i\tau) = 1/2 - i\tau = 1/2 + i(-\tau)$. The completed function ξ and every quantity derived from it by the archimedean data inherit this exact invariance: $\xi(1/2 + i\tau)$ is even in τ , and the logarithmic-derivative identity of Appendix C expresses $\text{Re}[-(\zeta'/\zeta)(1/2 + i\tau)]$ entirely through $\psi(1/4 + i\tau/2)$ and reflection-symmetric polar terms, all even in τ after the symmetrization the functional equation enforces. The Fourier transform of an even function of τ is an even function of u , so any identity $\mathcal{F}$ of this class constrains only F_{even} . The odd component F_{odd} is, by definition, the part of F changing sign under $u \leftrightarrow -u$; an even constraint places no condition on it, since adding any odd function to a solution of an even constraint yields another solution. Formally, the map that sends F to $F + g$ for arbitrary odd g preserves every even constraint and alters F_{odd} , so the even constraints do not determine F_{odd} . The closure Θ is the determination of F_{odd} via the relation $F_{\text{odd}} = -F_{\text{even}}$ on the negative half-line; this relation is not itself an even constraint, since it equates an odd quantity to the negative of an

even one and thereby fixes the odd channel, and it is therefore not in the class generated by $\mathcal{T}$. Hence no $\mathcal{T}$ supplies Θ . ■

This is the robust reason, and it does not depend on the refinement of the functional-equation argument. Any identity in the parity-even class generated by the functional equation and its archimedean data, however sharp, constrains only the even part of the boundary distribution and is silent on the odd part that carries the zero-location information. It is the analytic analogue of a relativization barrier: just as a counting-type argument that survives the addition of an arbitrary oracle cannot decide a question whose answer flips across oracles, a parity-even identity that survives the addition of an arbitrary odd function cannot decide a question whose answer lives in the odd channel. The analogy is exact in its logic, not a metaphor.

A magnitude bound of Lindelöf type falls inside this wall, not outside it.

The Lindelöf hypothesis, $|\zeta(1/2 + it)| \ll t^\epsilon$, is frequently named as an independent route toward the location of the zeros. It is not an independent route, because the magnitude $|\zeta(1/2 + i\tau)|$ is even in τ : it is invariant under the conjugation that the reflection induces, so it is a parity-even quantity. By Theorem 7.1 any bound on it, however sharp, constrains only F_{even} and is silent on F_{odd} . A bound on the magnitude does not become a bound on the imaginary part of $-\zeta'/\zeta$, which is the odd-channel object Θ requires. The Lindelöf route therefore lies inside the even-side wall: it refines the channel the functional equation already controls and does not enter the channel where the hypothesis lives. This corrects a common mislisting and tightens, rather than weakens, the bracket: one of the routes usually offered as a way around the even-side wall is in fact a motion within it.

The even-side wall, so stated, is usually read as a barrier against a family of techniques, in the manner of a relativization obstruction: a class of methods is shown not to suffice, and it remains open that a method outside the class might. That reading understates what is present. The inaccessibility of the odd channel to the functional equation is not a contingent limitation of a technique family. It is a conservation law, and it follows from the spectral theorem applied to the functional equation regarded as an involution. We make this precise, because the conservation form is exact where the barrier form is merely suggestive, and because it identifies with no slack the single additional ingredient any closure must supply.

The substitution $R : s \mapsto 1 - s$ is an involution, R^2 being the identity, and the completed zeta function satisfies $\xi \circ R = \xi$, so R is a genuine symmetry of the object whose zeros are in question. Restricted to the critical line, $R(1/2 + i\tau) = 1/2 - i\tau$, so on the line R is exactly the reflection $\tau \mapsto -\tau$. The boundary trace of the functional-equation symmetry is the parity reflection in the line coordinate, not an analogue of it. Any involution splits the space it acts on into a $+1$ eigenspace and a -1 eigenspace, here the even and the odd tempered

distributions in τ ; the parity decomposition of Section 4.3 is exactly this eigenspace decomposition. The parity of the boundary distribution is fixed by Schwarz reflection: because $\zeta(\bar{s}) = \zeta(s)^-$ for the real-coefficient Dirichlet series, $-(\zeta'/\zeta)(1/2 - i\tau) = -(\zeta'/\zeta)(1/2 + i\tau)^-$, so the real part of $-(\zeta'/\zeta)(1/2 + i\tau)$ is even in τ and the imaginary part is odd, the real part being the $+1$ eigenspace and the imaginary part the -1 eigenspace. This is a structural identity, verified to machine precision. By Theorem 4.2 and Section 4.3 the Riemann Hypothesis is the determination of F_{odd} , the imaginary part of $-\zeta'/\zeta$ on the line, and therefore lives entirely in the -1 eigenspace of the functional-equation involution.

Theorem 7.2 (parity conservation). Let R be the functional-equation involution $s \mapsto 1 - s$, acting on the boundary distributions of the critical line as the reflection $\tau \mapsto -\tau$, with $+1$ and -1 eigenspaces V_+ and V_- , the even and odd tempered distributions in τ . Every quantity derivable from the functional equation alone, transported to the line, lies in V_+ . The Riemann Hypothesis is the determination of a specific nonzero quantity lying in V_- . Since R acts as the identity on V_+ and as multiplication by -1 on V_- , and eigenspaces of distinct eigenvalues meet only at the zero distribution, no quantity in V_+ determines any nonzero quantity in V_- . The functional equation is therefore conservative for parity: it carries no information into the -1 eigenspace, and the Riemann Hypothesis, living wholly there, cannot be closed by any consequence of the functional equation whatsoever.

Proof. The functional equation is the single relation $\xi \circ R = \xi$. Any identity derived from it is a consequence of the invariance of ξ under R , hence is itself R -invariant when transported to the line, hence lies in V_+ ; this is Theorem 7.1, and it covers the adjoined archimedean data because the Gamma factor, the digamma terms, and the polar terms at 0 and 1 are all R -invariant on the line. An operator acting on a space with eigenspaces V_+ and V_- for distinct eigenvalues cannot carry a nonzero element of one to a nonzero element of the other, and no element of V_+ equals a nonzero element of V_- since the eigenspaces meet only at zero. The determination of F_{odd} is the determination of a specific nonzero element of V_- ; no R -invariant quantity, lying in V_+ , can supply it; the functional equation supplies only R -invariant quantities; hence the functional equation cannot supply F_{odd} , and by Theorem 6.2 cannot close the Riemann Hypothesis. The obstruction is exact: not that the even data is insufficient information about the odd sector, but that it is provably no information about it, by the orthogonality of the eigenspaces of an involution. ■

One objection meets this theorem directly and must be answered, because the answer is what fixes its strength. The closure relation is $F_{\text{odd}}(u) = -F_{\text{even}}(u)$ for $u < 0$, and F_{even} is R -invariant and is supplied by the functional equation; it may therefore seem that the even data, through this relation, determines the odd part on the negative half-line. The appearance is inverted. The relation $F_{\text{odd}} = -F_{\text{even}}$ on the negative half-line is not a consequence of the functional

equation; it is the Riemann Hypothesis itself, by Theorem 6.2. The functional equation supplies the value of F_{even} on the whole line. It does not supply the relation tying F_{odd} to F_{even} below zero, because that relation is the proposition in question. Knowing the $+1$ eigenspace in full says nothing about whether the -1 eigenspace equals its negation on half the line; that equality is an additional fact of odd-sector content, which the symmetry conserving parity cannot certify. The objection, pressed to the end, is the observation that one who already had the closure relation would have the odd part, which is true and is the definition of the closure, and is the reason the closure is the hypothesis rather than a corollary of the even data. The relation is the very thing the theorem shows the functional equation cannot reach.

A second objection concerns the exhaustiveness of the parity dichotomy and must be met with equal care, since the eigenspace decomposition is the engine of the theorem. One might grant that V_+ and V_- are the eigenspaces of R and still ask whether some admissible analytic constraint is parity-mixed, neither even nor odd, and so escapes the dichotomy, lying in neither eigenspace and therefore neither barred by the wall nor counted in the closure requirement. The decomposition forecloses this. Every tempered distribution on the line decomposes uniquely as an even part plus an odd part, the projections onto V_+ and V_- , with no third summand; the two eigenspaces span the whole space, and a parity-mixed object is precisely a nonzero sum of a V_+ part and a V_- part, not an element outside both. A constraint that determines F_{odd} does so through its V_- projection, by the definition of the odd component, and that projection is odd-sector content. The content of Theorem 7.2 is therefore exactly this: the functional equation supplies only V_+ content, the determination of F_{odd} is V_- content, and any constraint that supplies the latter must carry a nonzero V_- projection that the functional equation does not and cannot provide, whether that constraint presents itself as a symmetry, a positivity, or a one-sided growth bound. A parity-mixed constraint does not evade the theorem; its bearing on the hypothesis is carried entirely by its odd projection, and that projection is the content the functional equation conserves itself away from.

The upgrade over the barrier reading is in kind. A relativization-style barrier leaves open that a technique outside the family might suffice. The conservation theorem says the symmetry on which the entire functional-equation method rests is, by the spectral theorem, structurally silent on the eigenspace where the Riemann Hypothesis lives, so that no refinement, extension, or consequence of that symmetry can close the question, for the same reason an operator cannot transfer a value between its own orthogonal eigenspaces. This is the analytic analogue of a partition result that separates two classes not by an estimate but by establishing them categorically distinct, with the rigidity of the partition as the deliverable; here the rigidity is a conservation law of the only structural symmetry the zeta function is known to possess. What it does not do is close the hypothesis, and the reason marks exactly where the analogy holds and where it

stops. The Riemann Hypothesis is not the assertion that the even and odd channels are distinct, which the eigenspace decomposition makes immediate; it is the assertion that the odd channel takes one specific value. The conservation theorem establishes that the odd channel is free of the even channel; it does not establish what the odd channel equals. It proves the question lives in the -1 eigenspace and that the functional equation cannot enter that eigenspace; it does not evaluate the eigenspace.

The contribution is the resulting sharpening of the closure requirement. Closure requires an input independent of the functional equation that carries a nonzero odd-sector projection and forces F_{odd} to its critical-line value, the functional equation supplying, provably, none. In the one case where the analogous closure is a proven theorem, the function-field case, this input is a genuine second symmetry: the zeta function of a curve over a finite field possesses, beyond its functional equation, the action of Frobenius on the cohomology of the curve, which acts on the odd sector and forces the analogue of the Riemann Hypothesis through the positivity of the intersection form. The function-field zeta has two symmetries and the second closes the question; the Riemann zeta is known to possess only the first. We therefore name the missing ingredient a second symmetry after that precedent, while stating exactly what the theorem compels and what it does not: it compels that the closing input carry odd-sector content the functional equation cannot supply, and it leaves open whether that input is, for the classical zeta, a symmetry in the strict sense or a positivity or growth constraint carrying the requisite odd-sector projection. The two obstructions identified in Section 13, the absence of a constructed arithmetic surface over the integers and the wrong-signed arithmetic Hodge index of the structure that has been constructed, are precisely the obstructions to exhibiting the second symmetry in the form the function-field precedent realizes. The closure requirement of Section 12 is thereby restated in its sharpest form: not merely an odd-channel identity not reducible to the hypothesis, but an independent input supplying odd-sector content to the eigenspace the functional equation conserves. This is theorem-grade as a necessary condition and is matched to a proven precedent; whether such an input exists for the Riemann zeta function is open, and the theorem makes no claim that it does. It identifies, exactly, what closure requires and why the functional equation alone provably cannot supply it.

7.2 The Odd-Side Wall: the de Branges Positivity Overshoots into Falsity

The even-side wall shows the functional equation cannot enter the odd channel. The natural question is whether the structure built precisely to enter the odd channel reaches Θ . It does not, and the reason is the second wall.

The de Branges theory of Hilbert spaces of entire functions is the natural home for the odd channel. A de Branges space $\mathcal{H}(E)$ is built from a Hermite-Biehler function E , one satisfying $|E(\bar{z})| < |E(z)|$ for z in the upper half-plane; this defining inequality is a statement about the asymmetry of E under conjugation, and conjugation on the critical line is exactly the reflection $\tau \leftrightarrow -\tau$ that separates even from odd. The de Branges structure is therefore genuinely odd-channel: unlike the functional equation, it constrains the conjugation-antisymmetric content, the content Theorem 7.1 leaves free. De Branges proposed, in this setting, a family of positivity conditions whose validity would imply the Riemann Hypothesis.

Theorem 7.4 (odd-side wall, Conrey-Li). The de Branges positivity conditions that would imply the Riemann Hypothesis are not satisfied by the defining functions of the reproducing-kernel Hilbert spaces associated with the Riemann zeta function. The condition the de Branges approach requires is therefore strictly stronger than the Riemann Hypothesis and is false for the zeta function: the positivity that would close the question fails.

Attribution and proof. This is the theorem of Conrey and Li (2000; first circulated 1998). They exhibit explicit defining functions of the reproducing-kernel Hilbert spaces attached to ζ and to the Dirichlet L-function $L(s, \chi_4)$ and show that the de Branges positivity conditions fail for them; an argument communicated by Sarnak, reproduced in their paper, establishes the failure for the relevant space with no numerical computation. The de Branges conditions imply the generalized Riemann Hypothesis, so they are at least as strong as it; the Conrey-Li result shows they are strictly stronger, since they fail while the hypothesis is not known to fail, and in failing they remove the de Branges positivity as a route. We take the theorem as an external input, cited and not reproved, exactly as the embodied bounds of a physical argument are cited and not rederived. ■

The content of the odd-side wall is precise and is not the same as the even-side wall. The functional equation undershoots: it cannot reach the odd channel. The de Branges positivity overshoots: it enters the odd channel but demands more than the hypothesis, and that excess is false for the zeta function. The natural odd-channel construction does not land on Θ ; it lands past Θ , on a condition the zeta function refutes. Reaching Θ from the odd side therefore requires a construction that enters the odd channel without inheriting the de Branges over-demand, and no such construction is in hand.

7.3 The Bracket

The two walls together bracket Θ from opposite directions, and this is the consolidating structural fact.

Theorem 7.5 (the bracket). The odd-part closure Θ , which is the Riemann Hypothesis by Theorem 6.2, is bracketed between two theorem-grade failure-

directions. On the even side, every identity generated by the functional equation is parity-even and undershoots Θ , by Theorem 7.1. On the odd side, the natural positivity construction that enters the odd channel overshoots Θ into a false condition, by Theorem 7.4. The two standard channels miss Θ in opposite directions: the even channel cannot reach the odd-parity object, and the odd-channel positivity over-determines it into falsity. Any analytic route that closes the hypothesis must therefore do one of two things, both characterized in Scope 7.6: thread the gap between the walls, entering the odd channel without the even channel's parity limitation and without the natural odd construction's over-demand; or step outside the odd-side wall by building a modified odd-channel construction the Conrey-Li counterexample does not reach. The bracket walls off the two standard channels exactly; it does not claim to enclose a construction of the second kind, and it names that construction as one of the two open routes rather than absorbing it.

Proof. Immediate from Theorem 7.1 and Theorem 7.4. Θ is an odd-channel object by its definition as the determination of F_{odd} . The even toolkit constrains only F_{even} (Theorem 7.1), so it lies strictly on the near side of the odd channel: it does not reach Θ . The de Branges positivity constrains the odd channel but imposes a condition strictly stronger than the hypothesis that is false for ζ (Theorem 7.4), so it lies strictly on the far side: it overshoots Θ . The two enclosing conditions are of opposite parity-reach and opposite strength, and Θ is the object between them, in the odd channel and at the strength of the hypothesis exactly, neither reachable by the even-symmetric machinery nor delivered by the over-strong odd positivity. ■

This is the measurement at its sharpest. The literature records that the spectral approach has a domain ambiguity, that the analytic approach has not closed, and that the de Branges approach has a difficulty; each is recorded as a separate disappointment. The bracket replaces three disappointments with one structure. The obstruction is one object, that object is the hypothesis, and it sits in a named gap between a near wall and a far wall, each wall a theorem, the near wall proved here from parity and the far wall the published theorem of Conrey and Li. The gap is not a measure of how much remains to be tried; it is the exact location of the hypothesis between the two directions that provably miss it.

Scope 7.6 (what the bracket covers, and the two open routes distinguished). The two walls cover the two standard channels and no others. The even-side wall covers every identity generated by the functional equation and the archimedean data; the odd-side wall covers the de Branges positivity construction for the natural reproducing-kernel spaces of ζ . What lies outside both is a construction that enters the odd channel by genuinely independent arithmetic input without inheriting the de Branges over-demand. Two candidate sources remain visible, and they relate to the bracket geometry in two different ways, which we

distinguish precisely because the distinction is the exact boundary of what the bracket encloses.

The first route threads the gap between the walls. Automorphic input of Rankin-Selberg type, sharpened from averaged estimates to a pointwise statement on the odd channel, would enter the odd channel by independent arithmetic rather than by the parity-even symmetry, and would not be the de Branges positivity. Such an identity, if it determined F_{odd} at the strength of the hypothesis exactly, would land on Θ from inside the gap the two walls enclose: it threads between them. This route is genuinely interior to the bracket.

The second route steps outside the odd-side wall rather than threading the gap. A modified de Branges space, built from a different defining function, is not the natural construction the Conrey-Li theorem refutes; it is a different construction, and whether it overshoots Θ as the natural one does, or reaches Θ exactly, is open. We are explicit that the Conrey-Li theorem refutes the de Branges positivity for the natural ζ -space, not for every conceivable modified space. A successful modified construction would not thread the gap between the two named walls; it would go around the odd-side wall by changing the object the wall is stated about. The bracket does not enclose this route, and we do not claim it does. The bracket walls off the two standard channels; a construction that redefines the odd-side object is, by that redefinition, outside the bracket's scope, and it is one of the two open routes rather than a refinement absorbed by Corollary 7.7.

This is the precise boundary of the claim. The bracket encloses the two standard channels exactly, and localizes the hypothesis between them. It does not enclose the full space of possible constructions, since a modified odd-channel object lies outside the geometry the two walls define. The two routes are exhaustive as positions relative to the walls: a closing construction either lands on Θ from inside the gap the walls enclose, the interior thread, or reaches Θ by redefining the object the odd-side wall is stated about, the exterior step. This classification is by position, not by source. An odd-channel determination sourced from outside the prime-zero ledger, the operator presenting the zeros as its spectrum independently of ζ that Section 12 names, is not a third position relative to the walls but a source for the first: if such an operator supplied the odd-sector content at the strength of the hypothesis, it would land on Θ from inside the gap, threading between the walls by independent arithmetic exactly as the interior route requires. The enumeration here, over positions relative to the two walls, and the enumeration of Section 12, over sources of the closing input, are therefore consistent: every source, automorphic, external-operator, or other, that determines F_{odd} does so either by threading the gap or by stepping around the odd-side wall, and the external operator threads it. Any route that closes the hypothesis is the interior thread or the exterior step, and by Theorem 6.2 the difficulty of either is exactly the difficulty of the hypothesis.

Corollary 7.7 (permanence of the bracket). The bracket is stable under every refinement of the two standard channels. No sharpening of functional-equation methods supplies Θ , since each such advance produces a parity-even identity and Θ is odd, by Theorem 7.1. No sharpening of the de Branges positivity for the natural ζ -space supplies Θ , since that positivity is already false for the zeta function, by Theorem 7.4, and a sharper false condition is still false. The single object that stands between the unconditional even-part fact and the Riemann Hypothesis is therefore not reachable by improving either standard channel; it is reachable, if at all, only by a construction that threads the gap the two walls enclose. This is a positive and permanent structural fact, not a limitation discovered in passing. The persistence of the open problem over a hundred and sixty-five years is, on this reading, the structural consequence of two facts together: the even channel that built the problem cannot enter the channel where the answer lives, and the odd-channel construction built to enter it overshoots into falsity.

8. THE PRIME FIELD AT EQUILIBRIUM AND ITS DISCRIMINATING EXPONENTS

This section develops the theoretical-physics interpretation of the zeta function as the partition function of an idealized prime field, identifies the equilibrium universality class by four orthogonal critical exponents, and measures exactly which of those exponents discriminate the Riemann Hypothesis from its negation. The algebraic identification is exact; the universality picture is heuristic and inductive, and we maintain the distinction throughout.

8.1 The Partition-Function Identification

Define $Z(s) = \sum_n n^{-s} = \zeta(s)$ for $\text{Re}(s) > 1$, and identify $\zeta(s)$ as the partition function of an idealized prime field at inverse temperature $\text{Re}(s)$, with each prime p an elementary excitation of energy $\log p$, occupation numbers $k = 0, 1, 2, \dots$ the prime powers, and an integer $n = \prod p^{\{a\}}$ a multi-mode excitation of energy $\log n$. The Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1} = \prod_p \sum_k p^{-ks}$ is the exact factorization over independent prime modes, each factor the Bose-Einstein-type sum over occupations. The identification is an algebraic equality, not an analogy. The free energy is $F(s) = -\log \zeta(s)$, and its first derivative, the response function, is $\zeta'(s)/\zeta(s) = -\sum_n \Lambda(n) n^{-s}$, so $-\zeta'/\zeta$ is the negative response function. The boundary distribution central to Sections 3 through 7 is therefore the response function of the prime field on the critical line.

8.2 The Critical Line as Equilibrium Line

For $\text{Re}(s) > 1$ the series converges absolutely, the high-temperature ordered phase; for $\text{Re}(s) < 0$ the functional equation provides the dual ordered phase under $s \leftrightarrow 1 - s$ with Gamma-factor inversion; between them lies the critical strip, and the critical line $\text{Re}(s) = 1/2$ is its exact midpoint and the unique fixed-point set of the involution $s \leftrightarrow 1 - s$. Physically the critical line is the equilibrium line where the two ordered phases meet. In equilibrium critical phenomena, the singular structures of the partition function, the zeros, are predicted by the renormalization-group fixed-point structure to localize at the equilibrium boundary. The Riemann Hypothesis is, in this language, the assertion that the zeros lie precisely on the equilibrium line, the natural prediction of the framework.

8.3 Four Orthogonal Critical Exponents

Four exponent measurements are available, each established unconditionally as a proved theorem or directly verified fact, each targeting a distinct functional property. The word orthogonal here denotes that the four measure four distinct

functional properties of the prime field, not that their bearing on the hypothesis is statistically independent. As Proposition 8.1 makes precise, two of the four, the direct zero localization and the prime-counting envelope, both respond to the same underlying quantity, the deviation of an off-line zero from the critical line, and are therefore perfectly correlated in their bearing on the hypothesis above any counterexample height. They are distinct measurements of one discriminating source, not independent discriminating axes, and we group them as the two discriminating exponents in Section 8.4 accordingly.

Exponent Z_1 , Selberg variance. Selberg's theorem gives $\log|\zeta(1/2 + it)| / \sqrt{((1/2) \log \log T)} \rightarrow N(0,1)$ in distribution for $t \in [T, 2T]$, so the order-parameter logarithm has variance $(1/2) \log \log T$ on the critical line. Variance growing as $\log \log T$ rather than as a power of T is the fluctuation signature of the Gaussian class on the critical line; on any other vertical line in the strip the variance scales differently. The measurement is unconditional, a probabilistic statement about $|\zeta|$ requiring no hypothesis on zeros.

Exponent Z_2 , GUE level repulsion. The Montgomery-Odlyzko correspondence gives the rescaled pair correlation of consecutive zeros converging to the Gaussian Unitary Ensemble form $R_2(\alpha) = 1 - (\sin \pi\alpha / \pi\alpha)^2$. The level-repulsion exponent of GUE is $\beta = 2$, the universality signature of self-adjoint Hamiltonians with broken time-reversal symmetry, distinguished from the orthogonal class $\beta = 1$ and the symplectic class $\beta = 4$. Montgomery's convergence is conditional for restricted test functions, but Odlyzko's numerical verification over millions of zeros to heights of order 10^{22} is direct empirical fact, independent of any hypothesis.

Exponent Z_3 , direct zero localization. Rigorous computation has verified the hypothesis for all zeros up to height 3×10^{12} (Platt and Trudgian, 2021), every located zero lying on the critical line to arithmetic precision, with no off-line zero found. This is the order-parameter localization exponent, measured to be zero deviation in the verified range. The inference from no off-line zero up to that height to no off-line zero at any height is inductive; the measured exponent is zero deviation, with strong but not unconditional implication for the population. The quantity verified is a height, the bound on the imaginary part of the zeros checked, not a fixed count, and the exponent's force is bounded to that computed height however the height is expressed.

Exponent Z_4 , prime counting envelope. The error term $\Delta(x) = \pi(x) - \text{Li}(x)$ has been measured for x up to order 10^{22} , with $|\Delta(x)|$ within the envelope $C \sqrt{x} \log x$ consistent with the Riemann-Hypothesis-predicted bound. Off-line zeros at $\beta > 1/2$ would force deviations of order x^β exceeding the envelope by $x^{\{\beta - 1/2\}}$; none has been observed at any computed height. The $\sqrt{x} \log x$ scaling is the equilibrium response-amplitude exponent, measured consistently across many decades.

8.4 Which Exponents Discriminate the Hypothesis

We now state precisely which of the four exponents carry information about the truth of the Riemann Hypothesis and which are truth-invariant across it, since an inflated reading would treat all four as evidence of equal bearing. The treatment is the information-theoretic one: an exponent discriminates the hypothesis if its value differs between the world where the hypothesis holds and the world where it fails but no counterexample has yet been found; it is non-discriminating if its value is identical across the two and therefore carries zero mutual information with the truth.

Proposition 8.1 (discrimination profile of the four exponents). Let H denote the indicator of the Riemann Hypothesis, $H = 1$ in the world where it holds and $H = 0$ in the world where it fails with the lowest counterexample above all computed heights. Among the four exponents, Z_1 (Selberg variance) and Z_2 (GUE repulsion) are truth-invariant across H and carry zero mutual information with it, while Z_3 (zero localization) and Z_4 (prime counting envelope) are discriminating, their measured values being consequences that a counterexample would eventually violate:

$$I(Z_1 ; H) = 0, I(Z_2 ; H) = 0, I(Z_3 ; H) > 0, I(Z_4 ; H) > 0,$$

the last two evaluated at heights below any counterexample, where the measured values are equally consistent with both hypotheses, so that the present finite-sample measurements of Z_3 and Z_4 confirm the hypothesis only up to the computed height and not beyond.

Proof. The Selberg variance Z_1 is a statement about the distribution of $\log|\zeta(1/2 + it)|$ on the critical line, established unconditionally and identically in both worlds: Selberg's theorem makes no use of the location of zeros off the line and holds whether or not such zeros exist, so the value of Z_1 is the same under $H = 0$ and $H = 1$, and $I(Z_1 ; H) = H(Z_1) - H(Z_1 | H) = 0$ by the same collapse of conditional to marginal entropy used in the complexity-class setting. The GUE repulsion exponent Z_2 is a local-statistics property of the zeros that are present, on or off the line; broken-time-reversal level repulsion is a feature of the spectral correlations and is observed in the computed zeros regardless of whether a distant off-line zero exists, so Z_2 too is truth-invariant and $I(Z_2 ; H) = 0$. By contrast, Z_3 is the direct deviation of located zeros from the critical line; a counterexample is by definition a nonzero value of this deviation, so the exponent's value differs between the worlds and $I(Z_3 ; H) > 0$, with the caveat that below any counterexample height the measured deviation is zero in both worlds, so the discrimination is realized only asymptotically. Likewise Z_4 is the prime-counting envelope, which a counterexample would inflate by $x^{\beta - 1/2}$ at sufficiently large x ; the envelope's value differs between the worlds at large x and $I(Z_4 ; H) > 0$, again realized only above the counterexample height. ■

The universality-class identification therefore rests on a mixture of truth-invariant and discriminating exponents, and its evidential force is exactly delimited by this profile. The two truth-invariant exponents, Selberg variance and GUE repulsion, fix the universality class, the broken-time-reversal self-adjoint class, but say nothing about whether the hypothesis holds, since they take the same value in both worlds. The two discriminating exponents, zero localization and prime-counting envelope, do bear on the hypothesis, but only up to the computed height, since below any counterexample their measured values are also identical across the worlds. The universality argument's contribution is thus to identify the class within which the spectral operator has real spectrum, a strong structural picture in which the hypothesis is the natural prediction, while the two exponents that could in principle refute the hypothesis have done so only up to a finite height and cannot, by their finite-sample nature, establish it.

8.5 The Heuristic Status of the Universality Argument

The universality argument supplies strong structural evidence and does not bind mathematical truth. Exponent universality is an empirical regularity confirmed across physical systems by experiment and renormalization-group calculation, not a theorem about arithmetic objects, and its transfer to the prime field is a structural conjecture. Universality predicts pointwise zero localization as the typical class behavior, valid up to non-universal corrections that are subleading at large scale but nonzero at finite scale. Even granting pointwise exactness, the universality identification does not produce the odd-channel identity that Section 7 proves the two standard channels cannot supply; the two truth-invariant exponents fix the class but leave the odd-part closure Θ undetermined, and the two discriminating exponents are finite-sample. Any inference from physical universality to mathematical truth crosses the boundary from inductive generalization to deductive consequence, and the two registers are not interchangeable. The universality analysis is offered as theoretical physics on the prime field, articulating a coherent picture in which the hypothesis is natural, not as a proof.

8.6 An Independent Structural Witness: the Prime-Free Periodic Table

One further structural fact, drawn from atomic physics rather than from the prime field, witnesses the sector-separation that the bracket of Section 7 depends on. It is recorded here as an independent corroboration, not as a step in any proof.

The integers carry two distinct structures: an additive structure, the evenly spaced sequence under rotation and translation, and a multiplicative structure, the primes and their products. The Riemann Hypothesis is a statement about the

multiplicative structure; the entire analysis of this paper lives on the multiplicative Hilbert space $L^2(\mathbb{R}^{+*}, dx/x)$, and the obstruction Θ lives in the odd, conjugation-antisymmetric channel of that space. The question is whether the multiplicative-prime channel is genuinely a distinct sector from the additive channel, or whether the additive structure could reach it. Atomic physics supplies a clean answer.

The periodic table is a physical spectrum whose structure is set by spatial rotational symmetry and an integer counting index, with no reference to the primes. Its shell capacities are $2(2l + 1)$, forced by the representation theory of the rotation group $SO(3)$, the symmetry of physical space in three dimensions, together with the exclusion principle; its filling order follows the Madelung $n + l$ rule, an effective-theory consequence of the screened central-field problem rather than a theorem of $SO(3)$ alone, and it has the well-known exceptions of the transition and rare-earth elements, chromium and copper among them, where exchange and relativistic corrections shift the ground configuration; its period closures fall at the atomic numbers 2, 10, 18, 36, 54, 86, 118, fixed by cumulative filling under that rule. The two ingredients are spatial and arithmetic but not multiplicative: the angular structure is fixed by the rotation group of space, and the ordering parameter is the atomic number Z , the integer count of unit charges in the nucleus. The shell capacities are forced by the spatial rotation group; the filling order is an effective rule with exceptions; the ordering runs over the integers by the plain counting of Z ; and nowhere in the determination of the periodic table does the factorization of those integers into primes appear. The structure that fixes the chemistry of the elements uses spatial rotational symmetry and the integer counting of charge, and it has no need of the multiplicative structure of the integers at all.

This is the witness. The place where spatial rotational symmetry and the integer counting of charge set a physical spectrum, with its shell structure forced by the rotation group and its filling order an effective rule, is exactly the place that never invokes the primes. The multiplicative prime channel, in which the Riemann Hypothesis lives, is therefore a structurally distinct sector from the additive-and-counting channel that suffices to build the periodic table. The sector-separation the bracket relies on, between the machinery that the periodic table exemplifies and the multiplicative channel where Θ sits, is not an artifact of the framing of this paper; it appears independently in the architecture of matter, where the structure of the elements is fixed without the factorization of the integers ever entering. The periodic table is the worked example of a physical spectrum built from spatial symmetry and integer counting alone, and it is built that way precisely because it never enters the multiplicative channel where the hypothesis lives. We hold this as corroboration and not as a step in any proof; we do not rest it on a claim that the filling order is forced by symmetry, which it is not, and we do not claim that the additive structure of the integers generates the

spatial rotation group, which it does not. The single load-bearing observation is the absence of the primes, and that absence is what the witness records.

9. THE MULTI-AXIS STRUCTURAL DETERMINATION

The Riemann Hypothesis concerns an object instantiated at once in several mutually irreducible registers: the analytic register of the meromorphically continued Dirichlet series, the arithmetic-topological register of intrinsic invariants of the integers, the thermodynamic register of the Euler product as exact partition-function factorization, and the universality register of an equilibrium critical class. A property of an object so instantiated is constrained from each register simultaneously, and the convergence of those constraints is a structural fact about the object when the constraints are genuinely independent. This section states the four constraining axes, verifies their independence by three explicit conditions, identifies the single shared quantity whose removal would dissolve the convergence, and gives the determination in the two registers in which the hypothesis has a status, the strict formal register of syntactic proof and the physical-structural register of convergent determination.

9.1 The Four Constraining Axes

The first axis is formal. By Theorem 4.2 the hypothesis is equivalent to membership of $-(\zeta'/\zeta)(1/2 + it)$ in the Hardy space of the lower half-plane, equivalently to causal Fourier support on $[0, +\infty)$, the prime-power atoms beginning at $\log 2$ under the hypothesis. The axis is stated entirely in the vocabulary of functional analysis and distribution theory. It supplies the precise analytic target the other axes bear upon; it does not by itself close, the obstruction of Section 4.4 standing between it and a proof.

The second axis is empirical and thermodynamic. By the partition-function identification of Section 8.1 and the four exponents of Section 8.3, the prime field on the critical line lies in the equilibrium class of self-adjoint dynamics with broken time-reversal symmetry: the Selberg variance fixes the fluctuation exponent, the Montgomery-Odlyzko correspondence the symmetry-class exponent $\beta = 2$, the rigorous verification the localization exponent to height 3×10^{12} , and the prime-counting envelope the response-amplitude exponent. The axis is stated in the vocabulary of equilibrium statistical mechanics and random-matrix theory. Within this class the spectral operator carries real spectrum.

The third axis is arithmetic-topological, three intrinsic invariants of the integers that bear on the location independently of any analytic operation on ζ . The first is the causal support of the von Mangoldt measure. The measure $d\mu(x) = \sum_{\{p,k\}} \Lambda(p^k) \delta(x - p^k)$, carried to the logarithmic coordinate $u = \log x$, is supported on the set $\{k \log p : p \text{ prime}, k \geq 1\}$, whose smallest element is $\log 2$, the logarithm of the smallest prime, with no support below $\log 2$. This is a fact about which integers are prime powers, an invariant of $\mathbb{Z}$ under multiplication,

and its Mellin transform is exactly $-\zeta'/\zeta$; the lower bound $\log 2$ on the support is the arithmetic precondition for the causal threshold in the Hardy-space target of the first axis. The connection runs one way only: the bounded-below support of the measure in the half-plane of absolute convergence is the precondition for the causal threshold, and whether that support survives the meromorphic continuation to the critical line is exactly the open question of Section 4.4, not a consequence of the invariant. The second invariant is the critical line as phase boundary. The line $\text{Re}(s) = 1/2$ is the unique fixed-point set of the involution $s \leftrightarrow 1 - s$, the codimension-one boundary between the two stable phases of the field, fixed by the form of the functional equation and so by the multiplicative structure of $\mathbb{Z}$. The third invariant is arithmetic self-adjointness. The dilation group of the multiplicative integers acts unitarily on the multiplicative Haar space with self-adjoint generator H' by Section 3, the self-adjointness a property of the natural space the prime distribution inhabits rather than an imposed condition. The axis is stated in the vocabulary of measure theory, topology, and representation theory.

The fourth axis is the universal physics of order-parameter localization at continuous phase transitions. In the renormalization-group account of critical phenomena, the singular structures of the order parameter localize at the phase boundary as a consequence of the fixed-point structure, a regularity exhibited across physical systems: the Lee-Yang zeros of the Ising model on the imaginary-field axis, the Yang-Lee fugacity zeros on the unit circle, the chiral-condensate boundary in gauge theory. Applied to the prime field as a member of the class identified on the second axis, with phase boundary the critical line by the second invariant of the third axis, the principle places the singular structures of the partition function, the zeros of ζ , on the critical line. The axis is stated in the vocabulary of universal critical phenomena, and it is an empirical regularity established in specific model systems, not a general theorem that every partition-function zero of every system lies on its boundary at finite size; we hold it at that strength.

9.2 The Three Independence Conditions

The convergence of four axes is a structural fact only if the axes are genuinely independent, and three conditions make the independence explicit.

Vocabulary independence holds by direct inspection. The first axis is stated in functional analysis, the second in statistical mechanics and random-matrix theory, the third in measure theory and representation theory, the fourth in universal critical phenomena, and no axis borrows a term from another. The convergence is therefore not a single claim restated in four notations.

Mutual-information independence holds by examination of the inferential structure. No axis is derivable from the union of the others by inference native to its own register. The Hardy-space equivalence is not entailed by the universality

class, which constrains the spectrum to be real but does not supply the distributional formulation; the empirical exponents are independently proved theorems and directly verified measurements, not consequences of the formal axis; the arithmetic invariants are determined by the integers themselves; the universal principle is established in systems unrelated to the prime field. The four constitute genuinely distinct content.

Latent-covariate independence is the operative test, and it is the one that locates the open dimension. A latent covariate is a single quantity shared across axes whose removal would dissolve the convergence simultaneously, and three candidates carry across more than one axis. The zeta function itself is invoked on every axis; removing it, the second axis retains its content since the four exponents are measurements of the zero-spacings and the prime counts accessible without writing ζ , the third axis retains the von Mangoldt support and the involution fixed-point and the Haar self-adjointness as facts about the integers, and the fourth retains the universal principle as a result of statistical field theory, so the convergence survives. The functional equation is invoked on the first, third, and fourth axes; removing it weakens the even-part determination of the first axis but leaves the equivalence theorem standing, retains the empirical exponents measured at $\text{Re}(s) = 1/2$ without reference to it, and retains the von Mangoldt support and the universal principle, so the convergence survives. The prime number theorem is invoked in the bounds of the first axis and two exponents of the second; removing it weakens those and leaves the GUE statistics and the direct verification, the three arithmetic invariants, and the universal principle untouched, so the convergence survives. No one of these three dissolves the convergence, and to the extent the determination rests on them it is robust.

9.3 The Covariate That Does Not Subtract Out

One shared quantity remains, and it is the one that does not subtract out. The four axes converge on the location only through the identification of the spectrum of the self-adjoint operator H' with the imaginary parts of the zeros. The second axis forces the spectrum real and the fourth places the singular structures on the boundary, but each bears on the location of the zeros only if the zeros are that spectrum. This spectral correspondence is the quantity shared across the formal axis, which would read the zeros off the operator, and the empirical and universal axes, which constrain the operator. Subtract it, and the convergence opens: the first axis reverts to an equivalence with a condition not yet established, the second to a statement consistent with the location given the correspondence rather than forcing it, the fourth to a regularity exhibited in specific models. The correspondence is precisely the determination of the odd-part closure Θ . For the spectrum of H' to be the zeros is for the boundary distribution to carry the causal support that Θ names, and by Theorem 6.2 that is the hypothesis itself. The covariate that does not subtract out is therefore

identical to the obstruction of Section 6 and the open channel of Section 4.4. The convergence is fixed in the dimensions the four axes span and stands open in the one dimension that would close it to a proof, and that dimension is the odd-channel identity.

This closes a loop that any reader alert to circularity will look for, and we state the closure in plain terms rather than leave it implicit. Any verification apparatus applied to the four-axis convergence, including the one described in Appendix B, can return a nontrivial confirmation only by admitting the spectral correspondence as warrant, and the spectral correspondence is Θ , which is the hypothesis. Admit it, and the apparatus confirms consistency, but the discriminating content sits entirely in the admitted hypothesis. Withhold it, and there is no structural warrant left for the apparatus to act on, the only candidate being a quantity equivalent to the conclusion. Either way the apparatus contributes nothing to the determination that the convergence does not already contain, and we do not represent it as doing so. The apparatus of Appendix B is therefore load-bearing for no step here; the determination rests on the four axes and the single covariate that does not subtract out, and it points to the hypothesis without cashing the convergence as warrant.

9.4 The Determination in Two Registers

The hypothesis has a status in two registers, and the two are stated separately because they differ.

In the strict formal register, where a proof is a finite syntactic derivation, the hypothesis is not established. It reduces losslessly to the determination of one tempered distribution's odd part by Theorem 4.2 and Theorem 6.2; the even channel undershoots that determination by Theorem 7.1; the natural odd-channel positivity overshoots it into falsity by Theorem 7.4. The register returns the hypothesis open, localized to one object, and bracketed between two theorem-grade walls. This is the determination a reader who requires syntactic proof reads, and it is the spine of the paper.

In the physical-structural register, where a determination is read from the structure the object instantiates, the four axes converge on the critical line by four mutually independent constraints that survive removal of every shared covariate but one. The convergence is fixed in the dimensions the axes span: the equilibrium class is identified, the spectral arena carries real spectrum, the phase boundary is the critical line, the universal principle places the singular structures on that boundary, and the arithmetic support underwrites the analytic target. The one covariate that does not subtract out is the spectral correspondence, and it is identical to the strict-register gap. The two registers therefore meet on a single open object: the determination converges on the hypothesis as true, and the convergence stands pending the one identity that closes it, the determination of the odd-part component of $-\zeta'/\zeta$ on the critical

line. The physical-structural register returns a convergent determination that points to the hypothesis as true, pending the odd-channel identity.

The two registers do not compete and do not collapse into one word. The strict register withholds the proof and names the gap. The physical register supplies the convergent determination and names the same gap as its single open dimension. What is fixed is fixed in the dimensions where the evidence fixes it; what is open is named exactly, is identical across the two registers, and is the determination of one distribution's odd part. The hypothesis is, in the formal register, open and bracketed; in the physical-structural register, a convergent determination pointing to its truth pending that identity; and the two are the same structure read in two vocabularies.

10. THE CONSOLIDATED DETERMINATION

We now give the consolidated determination of the question of zero location under the single analytic discipline above. The question does not have one fate. It separates into layers that do not share a verdict, and the separation is the structural content of the result, since the layers that settle and the layer that stays open are divided by exactly the obstruction of Section 6, and naming the layers is naming where that obstruction sits. Each layer is stated at the grade its evidence supports.

The spectral arena seals. By Theorems 3.1 through 3.3, the Hilbert-Polya operator is uniquely fixed as H' on the multiplicative Haar space, self-adjoint with no boundary parameter, with real spectrum diagonalized by the Mellin transform. This layer is theorem-grade and unconditional. It certifies the arena in which the question is correctly posed; it does not place a zero. The deficiency-index ambiguity that forced prior approaches to external apparatus is dissolved.

The Cauchy anchor seals and is fenced. By Theorem 4.1, the Cauchy transform of $-\zeta'/\zeta$ vanishes identically in the safe regime to the right of the critical line, unconditionally and with no hypothesis on zeros. The fence is the content of Section 4.4: the anchor is established only for $\varepsilon > 1/2$, and the extension to the critical line is exactly what the three closure routes cannot supply without circularity. The layer seals as an unconditional fact about the safe regime; it does not reach the critical line.

The location is open and the obstruction is identified. Over the question of whether every zero lies on the critical line, the answer is not established. It leans toward the hypothesis on inductive grounds: the unconditional even-part determination consistent with it, the truth-invariant universality class within which the spectral operator has real spectrum, the two discriminating exponents confirmed to the computed height, and the field-wide expectation. The categorical proof is unavailable, and the reason it is unavailable is the content of Sections 6 and 7. The obstruction from the sealed layers to this open location is the odd-part closure Θ ; that closure is the hypothesis by Theorem 6.2; and it is bracketed between two theorem-grade walls, the even channel that undershoots it by Theorem 7.1 and the odd-channel de Branges positivity that overshoots it into falsity by Theorem 7.4. The even-side wall is moreover not a contingent barrier but a conservation law, by Theorem 7.2: the functional equation is an involution whose action on the critical line is the parity reflection, the hypothesis lives in the -1 eigenspace, and the symmetry carries provably no information into that eigenspace, so the unavailability of a functional-equation proof is structural and permanent rather than a limitation awaiting a better estimate. The location is therefore under-determined, with a direction, and bracketed on both sides; the direction is supplied by the structural picture, not by any closed argument; the two walls supply the exact failure-directions any closing argument

must avoid; and the conservation law identifies, as a necessary condition, the single ingredient a closing argument must add, a second symmetry acting nontrivially on the conserved eigenspace.

The contrary proposition lacks warrant of its own. The proposition that some non-trivial zero lies off the critical line raises no positive analytic warrant: no construction of such a zero, no computation exhibiting one up to the verified height of 3×10^{12} , no identity forcing one, and no structural argument requiring one. Its standing is the absence of warrant, not a demonstration of its falsity, since the location itself is open and an account that does not prove the hypothesis cannot, on pain of contradiction, prove its negation false. The asymmetry between the hypothesis and its contrary is an asymmetry of available warrant, the hypothesis carrying a sealed arena, a sealed anchor, and a directional lean, the contrary carrying nothing positive, and it is not an asymmetry of established truth value.

The consolidated determination is the following table.

Layer	Even-part / formal	Empirical / measured	Odd-part / location	Verdict
Spectral arena (operator H')	unique self-adjoint realization	-	-	sealed, theorem grade
Cauchy anchor ($\varepsilon > 1/2$)	unconditional vanishing	-	-	sealed, theorem grade, scope-fenced
Universality class	-	Z_1, Z_2 truth-invariant; Z_3, Z_4 discriminating to height	-	class identified; non-discriminating for the location
Periodic table (additive sector)	$SO(3)$ + Madelung fix the spectrum	closures at $Z = 2, 10, 18, 36, 54, 86, 118$	prime-free	additive sector complete without the prime channel
Location (RH)	even part fixed by FE	discriminating exponents to height	odd-part closure Θ undetermined	open, directional; obstruction = Θ
Some zero off the line	no warrant	no warrant	no warrant	absence of warrant, not falsity
$\Theta \Leftrightarrow RH$	parity identity	-	-	logical identity (Theorem 6.2)
Even channel reaches Θ	parity-even, undershoots	-	-	barred, even-side wall (Theorem 7.1)
Lindelöf reaches Θ	magnitude is parity-even	-	-	falls inside even-side wall, not a route
de Branges positivity reaches Θ	-	-	odd-channel, overshoots into falsity	barred, odd-side wall

Layer	Even-part / formal	Empirical / measured	Odd-part / location	Verdict
				(Theorem 7.4, Conrey-Li)
Θ bracketed	even undershoot	–	odd overshoot	bracketed both sides (Theorem 7.5)
Functional equation closes RH	involution, acts trivially on its +1 eigenspace	–	RH lives in the –1 eigenspace	impossible, parity conservation law (Theorem 7.2)
What closure requires	–	–	a second symmetry acting nontrivially on the –1 eigenspace	necessary condition (Theorem 7.2); realized in the function-field case by Frobenius on cohomology
Multi-axis convergence (four axes)	formal target	universality class, real spectrum	arithmetic support; spectral correspondence open	convergent determination pointing to RH true, pending a second symmetry on the conserved eigenspace

The table is read down the verdict column as one determination. The arena and the anchor seal at their honest grades and do not move with any refinement of method. The universality class is identified by two truth-invariant exponents and is silent on the location, while two discriminating exponents confirm the hypothesis to the computed height and no further. The periodic table witnesses, independently, that the additive sector of the integers fixes a full physical spectrum without ever entering the prime channel. The location is open with a direction, and it is reached by the analysis exactly up to one object, the odd-part closure Θ , which is the hypothesis by Theorem 6.2 and is bracketed between the even channel that undershoots it (Theorem 7.1) and the odd-channel positivity that overshoots it into falsity (Theorem 7.4). The contrary proposition breaks for want of any positive warrant, typed as absence of warrant rather than as established falsity. Two facts in the table carry the structural weight together.

The first is the bracket row: the hypothesis is reached by the unconditional machinery exactly up to one named object, that object is the hypothesis, and the two standard channels that could reach it miss in opposite directions, one too weak and one too strong-and-false. The second is the conservation row, which converts the even-side miss from a barrier into a law: the functional equation is an involution, the hypothesis lives in the eigenspace the involution acts on trivially, and by the spectral theorem no consequence of the functional equation can ever enter that eigenspace, so the persistence of the open problem is the structural signature of a conserved quantity and not a record of insufficient ingenuity. The two facts together yield the closure characterization of the next row: what the unconditional machinery cannot supply, and what a closing argument must therefore add, is a second symmetry acting nontrivially on the conserved eigenspace, exactly the ingredient that closes the proven function-field analogue through the action of Frobenius on cohomology. The final row reads the same structure from the physical-structural register: the four mutually independent axes converge on the critical line, the convergence is fixed in the dimensions the axes span, and its single open dimension, the spectral correspondence, is identical to the odd-part closure Θ of every row above. The two registers therefore agree on one open object. In the formal register the hypothesis is open and bracketed, with the even-side wall sharpened to a conservation law and the closure requirement sharpened to a second symmetry; in the physical-structural register it is a convergent determination pointing to its truth, pending that second symmetry on the conserved eigenspace.

11. THE CONSOLIDATED VERDICT

The determination, the universality profile, the periodic-table witness, and the consolidating theorems consolidate into a single picture. The spectral arena and the Cauchy anchor seal at theorem grade, the first dissolving the deficiency-index ambiguity, the second establishing an unconditional vanishing in the safe regime. The Riemann Hypothesis is reformulated losslessly as a causal-support condition on one tempered distribution, that condition is reduced by parity to the determination of a single odd-part object, and the consolidating contribution fixes the relation between that object and the hypothesis, then brackets it. The identity proves the odd-part closure logically equivalent to the hypothesis, so the obstruction is not adjacent to the problem but identical to it. The two-sided barrier proves the closure enclosed between two theorem-grade walls: the even channel, being symmetric under the critical-line reflection, undershoots the odd-parity object and cannot reach it, so no refinement of functional-equation methods supplies it; and the de Branges positivity, the natural construction that does enter the odd channel, overshoots it into a condition the zeta function fails to satisfy, by the theorem of Conrey and Li, so no sharpening of that construction for the natural ζ -space supplies it either. The even-side wall is sharpened to a conservation law: the functional equation is an involution whose action on the critical line is the parity reflection, the hypothesis lives in the eigenspace the involution fixes pointwise as its trivial eigenvalue acts elsewhere, and by the spectral theorem no consequence of the functional equation carries any information into the eigenspace where the hypothesis lives, so the even-side unavailability is permanent and structural rather than provisional. From the conservation law the closure requirement takes its sharpest form, as a necessary condition: a closing argument must supply a second symmetry of the zeta function, independent of the functional equation, acting nontrivially on the conserved eigenspace, which is the structure that closes the proven function-field analogue through the action of Frobenius on cohomology. The universality analysis identifies the equilibrium class by four exponents and measures which discriminate the hypothesis: two are truth-invariant and fix the class without bearing on the location, two are discriminating but finite-sample. The periodic table witnesses independently that the additive, rotation-symmetric sector of the integers fixes a full physical spectrum without invoking the primes, confirming the prime channel is a distinct sector. The contrary proposition breaks for absence of any positive warrant, typed as absence of warrant rather than as established falsity.

The consolidated picture is a measurement and a bracket. The distance from the unconditional Cauchy anchor to the critical line is one object wide; the object is the odd-part closure; the closure is the Riemann Hypothesis itself; and it is bracketed between two named, theorem-grade failure-directions, the even channel that falls short of it and the odd channel that overshoots it into falsity.

This is bolder than a claimed proof and more durable, because it cannot be overturned by a sharper estimate from either channel: the gap it names is the hypothesis, and the two walls that bracket it are theorems, one proved here from parity and one the published theorem of Conrey and Li. The bracket encloses the two standard channels exactly. A route that closes the hypothesis is either an interior thread between the two walls or an exterior step outside the odd-side wall by a modified construction the Conrey-Li counterexample does not reach, and Section 7 characterizes both; neither is the refinement of a standard channel that the field has expected, and that is the content of the localization. The proposition that the hypothesis holds stands strictly above the proposition that some zero lies off the line, the first carrying a sealed arena, a sealed anchor, a directional lean, and a two-sided localization, the second breaking for absence of any positive warrant. We make no claim of a proof of the hypothesis. Theorem 6.2 identifies the obstruction as the hypothesis; Theorem 7.1 walls it off from the even channel; Theorem 7.2 upgrades that wall to a conservation law of the functional-equation involution and identifies the second symmetry a closure must add; Theorem 7.4 walls it off from the natural odd-channel positivity; Theorem 7.5 brackets it between the two; the determination seals the arena and the anchor and leaves the location open with a direction. At no point is the hypothesis established. It remains open, localized to one object, bracketed between the two standard channels that provably miss it, and sharpened by the conservation law to the exact closure requirement of a second symmetry on the conserved eigenspace.

12. WHAT CLOSURE WOULD REQUIRE

For the reader who asks what would close the question, the answer is exact and follows from the bracket. Closure requires an analytic identity that determines the odd-part component F_{odd} of the boundary distribution, equivalently the imaginary part of $-\zeta'/\zeta$ on the critical line, and by the two-sided barrier that identity must thread between the two walls. It must enter the odd channel, which by Theorem 7.1 the functional equation and every parity-even refinement cannot do; and it must not inherit the de Branges over-demand, which by Theorem 7.4 is false for the natural ζ -space. By Theorem 7.2 this requirement takes its sharpest form: since the functional equation is conservative for parity and acts trivially on the eigenspace where the hypothesis lives, closure requires a second structural symmetry of the zeta function, independent of the functional equation, acting nontrivially on the -1 eigenspace and forcing F_{odd} to its critical-line value. This is exactly the structure that closes the proven function-field analogue, where the action of Frobenius on the cohomology of a curve is the second symmetry and the positivity of the intersection form is the force; the Riemann zeta is known to carry only the first symmetry, and Section 13 records the two obstructions to exhibiting the second. Two candidate sources remain visible, each a substantial open problem. The first is automorphic input of Rankin-Selberg type, sharpened from averaged estimates to a pointwise statement on the odd channel; this enters the odd channel by genuinely independent arithmetic rather than by the parity-even symmetry, and is not the de Branges positivity. The second is a modified de Branges space, built from a defining function for which the Conrey-Li counterexample does not apply; whether such a space evades the overshoot is open, and we are explicit that Conrey and Li refute the positivity for the natural reproducing-kernel space of ζ , not for every conceivable modification. Either, if produced, closes the hypothesis by Theorem 6.2, and the difficulty of producing either is, by the same theorem, exactly the difficulty of the hypothesis. Whether such an identity exists within the resources of classical analytic number theory, or requires resources of a different kind, is itself open. The structural perspective of this paper does not assert that closure is impossible; it asserts that closure must thread a named gap between an even-channel undershoot and an odd-channel overshoot, supply a second symmetry on the eigenspace the functional equation conserves, and it locates that gap precisely as the determination of one tempered distribution that the even symmetry cannot reach and the natural odd positivity over-demands. The contributions of the paper stand independent of which outcome obtains.

The three routes by which the odd channel has been approached in the literature each reduce, on inspection, to the hypothesis itself, and we record the reduction because it makes the closure requirement concrete rather than gestural. The first route is the argument-function moments. The imaginary part of the logarithm of ζ on the critical line is the argument $S(t)$, tied to the zero counting

by $N(t) = (t/2\pi) \log(t/2\pi e) + (1/\pi) S(t) + O(1)$, so the odd channel is exactly the carrier of the zero fluctuations. The sharp results on its distribution, the exponential-moment bounds of Najnudel (2018) and the third-moment estimates of Fazzari and Gerspach (2024), are established conditionally on the Riemann Hypothesis and on further pair- and triple-correlation conjectures; they describe the odd channel with precision given the hypothesis, and they do not determine it toward the hypothesis. The implication runs from the hypothesis to the channel, which is the wrong direction for closure.

The second route is the Speiser equivalence. Speiser's theorem, with the rigorous proof of Levinson and Montgomery (1974), establishes unconditionally that the Riemann Hypothesis holds if and only if the derivative $\zeta'(s)$ has no non-real zeros in $0 < \text{Re}(s) < 1/2$, a genuinely odd-channel reformulation, since the horizontal distribution of the derivative's zeros is governed by the argument structure rather than by the even magnitude. To close the hypothesis through it requires proving that zero-free region for ζ' , and the literature on the horizontal distribution of ζ' zeros near the line, from Levinson and Montgomery through Soundararajan, Zhang, Feng, and Garunkštis, leaves this open and identifies it as governed by the spacing of the zeros of ζ , which is the location the hypothesis fixes. That the route turns on the even functional equation not being sufficient is confirmed in the wild: the Davenport-Heilbronn function, an element of the extended Selberg class carrying a Riemann-type functional equation but possessing zeros off the critical line, does not satisfy the Speiser equivalent, exactly as the even-side wall of Theorem 7.1 requires, since a counterexample-bearing function with the same even structure shows the even symmetry cannot single out the on-line configuration.

The third route is the de Branges space, and its split is the sharpest confirmation of the bracket. The Hermite-Biehler structure-function property of the natural family $E_h(z) = \xi(1/2 + h - iz)$ holds unconditionally for $h \geq 1/2$ and, for the small- h spaces that reach the critical line, holds if the Riemann Hypothesis holds, as Lagarias (2006) records; this is the part of the de Branges apparatus that is downstream of the hypothesis, a consequence of it rather than a route to it, and it even supplies a self-adjoint operator with the zeros as spectrum once the hypothesis is granted. The part that would be upstream of the hypothesis, the positivity condition $\langle F(z), F(z + i) \rangle \geq 0$ that by de Branges's theorem would imply it, is the condition Conrey and Li (2000) refute for the defining functions of the reproducing-kernel spaces of ζ . The structure function is granted by the hypothesis; the positivity that would grant the hypothesis is false for the natural space. Whether a modified space built from a different defining function evades the Conrey-Li counterexample is open, and the recent work revising the de Branges axioms, including Bereza (2025), revises the axiomatic frame without exhibiting a working ζ -space; no evading construction has been produced.

The fourth route is the explicit formula and the prime-sum positivity built on it, and it is the route that states most plainly why the channel resists determination. The Guinand-Weil explicit formula is an identity equating a sum over the zeros to a sum over the prime powers, with the Gamma-factor archimedean term between them. From it Weil's positivity criterion and Li's criterion are derived, each a genuine equivalence: the Riemann Hypothesis holds if and only if the Weil quadratic form is non-negative on all test functions, and equivalently if and only if every Li coefficient λ_n is non-negative. Proving the positivity is proving the hypothesis, not a step toward it, and the positivity is not derivable from the prime side alone, because the explicit formula is a conserved ledger, not a one-sided constraint: it permits trading control of the zeros for control of the primes and back, and so it makes determination of the odd channel and location of the zeros two names for moving the same quantity. The same circularity disposes of the prime-pair variance form of the route. An attempt to force the hypothesis by showing the off-diagonal prime-pair sum cancels under Möbius weighting requires the Möbius sum to decay at the rate $M(x) = O(x^{\{1/2 + \varepsilon\}})$, and that decay rate is itself equivalent to the Riemann Hypothesis by the classical Mertens-type equivalence, so the cancellation that the argument needs is the hypothesis it would conclude. The explicit-formula route therefore does not escape the bracket; it is the cleanest statement of why the bracket holds, since it exhibits the zero side and the prime side as the two pans of one balance that no operation on a single pan can fix.

The four routes therefore agree, and their agreement corroborates the bracket from outside the paper's own theorems. The argument-moment route assumes the hypothesis to describe the odd channel; the Speiser route translates the hypothesis into an equally open odd-channel statement whose resolution is again governed by the zero locations; the de Branges route has its structure-function property downstream of the hypothesis and its closing positivity false for the natural space; the explicit-formula route is an identity whose positivity criteria are equivalences to the hypothesis and whose Möbius-decay form is the hypothesis restated. Each reduces to the hypothesis, which is what Theorem 6.2 predicts, since the odd-part determination is logically equivalent to the hypothesis and any honest route to it must already contain it. The closure requirement is thus not merely that some odd-channel identity be found, but that it be an identity not reducible in this way to the hypothesis it would establish, and no identity in the present literature meets that condition.

The agreement of the four routes is not accidental, and a single structural observation shows why no prime-side identity can meet the closure requirement, so that the absence of such an identity is a consequence of the conservation structure rather than a record of insufficient effort. The odd part of $-\zeta'/\zeta$ on the critical line is, pointwise, the regularization width of each pole. Near a zero $\rho = \beta + i\gamma$ the logarithmic derivative has a simple pole, and on the line $s = 1/2 + it$ the local contribution is $1/[(1/2 - \beta) + i(t - \gamma)]$, whose imaginary part is $-(t - \gamma)/$

$[(1/2 - \beta)^2 + (t - \gamma)^2]$. When $\beta = 1/2$ this is $-1/(t - \gamma)$, a divergent, purely imaginary, odd singularity at $t = \gamma$; when $\beta \neq 1/2$ the term $(1/2 - \beta)^2$ regularizes it into a finite peak of width $|1/2 - \beta|$. The odd channel is therefore the pointwise record of every deviation $\beta - 1/2$, and to determine the odd channel is to determine every $\beta - 1/2$, which is the hypothesis. This is the content of Theorem 6.2 displayed in the analytic shape of the channel rather than asserted.

From this the routes divide exhaustively into two, and the division is forced by the explicit formula being a conserved identity with the zero side on one pan and the prime side on the other. A candidate identity computed from the prime side is either insensitive to the deviations $\beta - 1/2$ or sensitive to them. If it is insensitive, it reads only the symmetric content and is true whether or not the hypothesis holds, the case of the odd test function in the explicit formula, whose zero-sum vanishes by the conjugate pairing $\gamma \leftrightarrow -\gamma$ that holds for any β , carrying no information about the location; such an identity is provable but truth-invariant across the hypothesis and does not close the channel. If it is sensitive to the deviations, then by the conservation of the identity its evaluation requires the zero side, the case of the second-moment integral whose finiteness on the line is the hypothesis itself, and of the signed asymmetry $\Sigma(\beta - 1/2)$ whose contour evaluation pushes the Dirichlet series from the region of absolute convergence down to the critical line across exactly the zeros whose deviations are sought; such an identity closes the channel but is the hypothesis restated. The two cases exhaust the prime-side identities, because the deviations live entirely on the zero pan of the balance, so a prime-side quantity that pinned them without their pinning it would be a function of one pan determining the other, which the conservation forbids. A prime-side odd-channel identity that is both independent of the hypothesis and sufficient to close it is therefore excluded, not unfound.

What this leaves open is exactly one kind of route, and it is one that does not lie on the prime-zero balance at all. A determination of the odd channel that is not computed from either pan of the explicit formula, but supplied by an external structure that exhibits the zeros as the spectrum of a self-adjoint operator given independently of ζ , would close the channel without standing on the conserved identity and so without reducing to the hypothesis through it. This is the Hilbert-Polya possibility in its strict form, an operator handed in from outside the arithmetic rather than constructed from the zeta function, and no such operator is known. The closure requirement, stated at its sharpest, is therefore that the odd-channel identity arrive from outside the prime-zero ledger, since every identity internal to that ledger is excluded by the conservation that makes the localization exact. The bracket is not a barrier awaiting a cleverer estimate from a standard channel; it is the visible form of a conservation law, and the only determination it does not foreclose is one sourced from a structure the ledger does not contain.

13. ASSUMPTIONS, GAPS, AND LIMITATIONS

We enumerate the assumptions and the open points, since a determination that does not state where it can fail is not yet a usable one.

The reformulation certifies an arena, not a location. The operator result of Section 3 fixes the unique self-adjoint realization on the multiplicative Haar space and dissolves the deficiency-index ambiguity. It does not place any zero on the critical line, and no reader should take the unique self-adjointness as bearing on the location; the location is fixed by the odd-part closure, which the operator does not supply.

The Cauchy anchor is unconditional only in the safe regime. Theorem 4.1 holds for $\varepsilon > 1/2$, where no zeros lie in the integration domain. The extension to the critical line is exactly the open problem, and the three closure routes are circular, as Section 4.4 exhibits by residue accounting.

The equivalence $\Theta \iff \text{RH}$ is exact but not deep. Theorem 6.2 is a parity reading of the causal-support condition and is near-immediate once Theorem 4.2 is in hand. We do not present it as a deep theorem; its value is the exactness of the localization, and the weight of the consolidating contribution rests on the two-sided barrier of Section 7.

The even-side wall is a conservation law, fenced to what it conserves. Theorem 7.1 proves the functional equation and its extensions parity-orthogonal to the odd-part closure, and Theorem 7.2 sharpens this from a barrier against a technique family to a conservation law of the functional-equation involution: by the spectral theorem the symmetry acts trivially on its own $+1$ eigenspace and the Riemann Hypothesis lives in the -1 eigenspace, so no consequence of the functional equation can close it. We are precise about the scope. This conserves what the functional equation cannot reach; it does not prove that no technique whatever can close the question. On the contrary, Theorem 7.2 identifies exactly the additional ingredient a closure must supply, a second symmetry acting nontrivially on the -1 eigenspace, and exhibits the proven function-field case as a realization of precisely that structure. The wall is held at its true strength, which is a conservation law of the only known symmetry of ζ , and is not extended to a claim that the conserved eigenspace cannot be reached by a symmetry the zeta function is not yet known to possess.

The odd-side wall is the published Conrey-Li theorem and is fenced to the natural ζ -space. Theorem 7.4 is cited, not reproved; it establishes that the de Branges positivity fails for the defining functions of the reproducing-kernel Hilbert spaces of the zeta function, refuting the de Branges route for that natural space. It does not establish that every conceivable modified de Branges space, built from a different defining function, must also overshoot; whether a modified

space evades the Conrey-Li counterexample is open, and Scope 7.6 states this explicitly. The odd-side wall is therefore a barrier against the natural odd-channel positivity construction, not against every possible odd-channel construction, and we do not extend it past the space Conrey and Li treat.

The bracket locates; it does not close. The two walls together bracket Θ between an undershoot and an overshoot, and this is a localization, not a proof. Bracketing the location of an object is not eliminating the alternatives to its truth: the bracket says where the hypothesis is and that the two standard channels miss it, and says nothing about whether the object at that location holds. No reader should take the bracket as implying the hypothesis is true; that inference does not follow and we do not draw it.

The universality argument is heuristic and finite-sample. The class identification of Section 8 rests on two truth-invariant exponents, which fix the class but do not bear on the location, and two discriminating exponents, which confirm the hypothesis only to the computed height. The transfer of universality from physical systems to the prime field is a structural conjecture, and the inference from class membership to pointwise zero location requires an exactness that universality alone does not provide.

The periodic-table witness is structural corroboration, not a step in any proof. Section 8.6 observes that the additive, rotation-symmetric sector of the integers fixes the periodic table without invoking the primes, corroborating that the prime channel is a distinct sector. This is an independent structural observation that strengthens the plausibility of the sector-separation; it is not a derivation of the barrier, which rests on the parity argument of Theorem 7.1 and the Conrey-Li theorem of Theorem 7.4, and it carries no weight in those theorems.

The discrimination profile is conditioned. Proposition 8.1 evaluates the discriminating exponents below any counterexample height, where the measured values are common to both worlds; the discrimination is realized only asymptotically, and the finite-sample measurements confirm the hypothesis only up to the computed height.

The multi-axis convergence is a determination, not a proof. The four-axis structure of Section 9 converges on the critical line by constraints that survive removal of every shared covariate but one, and that surviving convergence is the convergent determination of the physical-structural register. It is not a proof in the strict formal register. Three of its four axes are heuristic or finite-sample in part, the universality axis is an empirical regularity of specific model systems rather than a general theorem, and the convergence closes to a proof only when the one covariate that does not subtract out is supplied. We hold the multi-axis determination at exactly that grade: convergent, pointing to the hypothesis as true, and pending the closing identity, not establishing the hypothesis on its own.

The spectral correspondence does not subtract out, and that is the open dimension. Section 9.3 isolates the one covariate shared across the axes that survives every subtraction, the identification of the spectrum of H' with the imaginary parts of the zeros. That correspondence is the determination of the odd-part closure Θ , which by Theorem 6.2 is the hypothesis itself. The multi-axis convergence is therefore fixed in the dimensions its axes span and open in that single dimension, and the open dimension is identical to the obstruction of Section 6 and the gap of Section 4.4. No part of the multi-axis determination supplies that dimension, and we do not represent it as doing so. The openness of that dimension is structural and not a record of unfinished search. Section 12 establishes that the odd-part component is the pointwise regularization width of the poles of $-\zeta'/\zeta$ on the line, so its determination is the hypothesis, and that the explicit formula being a conserved identity excludes every prime-side identity that would determine it independently. The one route not foreclosed is a determination sourced from outside the prime-zero ledger, and we do not claim such a route or possess one.

No metaphysical claim is load-bearing. The body uses only the standard analytic theory of Section 4, the elementary operator theory of Section 3, the Conrey-Li theorem of Section 7, and the established critical-phenomena results of Section 8. The framework's interpretive commitments, including the reading of Appendix B and any monistic reading of the underlying structure, are confined to Appendix B and are used in no proof.

No claim on the hypothesis. We do not claim a proof of the Riemann Hypothesis. Theorem 6.2 identifies the obstruction as the hypothesis; Theorem 7.1 walls it off from the even channel; Theorem 7.4 walls it off from the natural odd-channel positivity; Theorem 7.5 brackets it between the two; the determination seals the arena and the anchor and leaves the location open with a directional lean. At no point is the hypothesis established, and it remains open, localized exactly and bracketed on both sides.

14. RELATED WORK

The spectral approach to the zeros originates with the Hilbert-Polya conjecture and is developed by Berry and Keating (1999) through the operator $H = xp$, whose domain ambiguity Section 3 resolves by the change to the multiplicative Haar geometry where Stone's theorem fixes the realization. Sierra and Townsend (2008) and Sierra and Rodríguez-Laguna (2011) addressed the same ambiguity through Landau levels and an xp -model regularization, each at the cost of external apparatus the present approach does not require. The work complements the adèle-class approach of Connes (1999) and Connes and Marcolli (2008) by isolating the archimedean component and showing it sufficient for the Hardy-space reformulation. The pair-correlation statistics of the zeros are due to Montgomery (1973) and Odlyzko (1987 and subsequent), extended to all correlations by Rudnick and Sarnak (1996) and embedded in random-matrix universality by Katz and Sarnak (1999) and Keating and Snaith (2000); the present paper reads these as the universality signature of the spectral operator and measures their discrimination profile against the hypothesis. The Selberg variance theorem is due to Selberg (1946). The rigorous verification of zeros is due to Platt and Trudgian (2021). The Hardy-space and Paley-Wiener-Schwartz machinery is standard (Reed and Simon, 1975; Titchmarsh, 1986). The de Branges theory of Hilbert spaces of entire functions, and the positivity conditions that would imply the Riemann Hypothesis, are due to de Branges (1986, 1992 and subsequent); the odd-side wall of Section 7 is the theorem of Conrey and Li (2000, first circulated 1998), with the no-numeric argument for the relevant space communicated by Sarnak and reproduced there, establishing that the de Branges positivity fails for the natural reproducing-kernel spaces of ζ . The even-odd parity decomposition of a causal-support condition is classical Fourier analysis; its application to localize the Riemann obstruction, the even-side parity barrier, and the assembly of the two-sided bracket from the parity barrier and the Conrey-Li theorem are the present contribution. The formalization of non-discrimination as vanishing mutual information follows the standard definition in Cover and Thomas (2006). The two-attractor architecture of the periodic table invoked in Section 8.6 (the rotational $SO(3)$ shell structure and the Madelung filling rule) is standard atomic physics. The verification operator's three-axis structure and its origin are described in Appendix B.

15. CONCLUSION

We have stated the operator reformulation at its earned strength, certifying a unique spectral arena and not the location of the zeros. We have supplied the unconditional Cauchy anchor in the safe regime and the lossless Hardy-space reformulation of the Riemann Hypothesis as a causal-support condition on one tempered distribution. We have proved the consolidating identity and assembled the two-sided barrier. The identity locates the obstruction, the odd-part closure, and proves it logically equivalent to the hypothesis, so that the obstruction is identical to the problem and not adjacent to it. The two-sided barrier encloses that obstruction between two theorem-grade walls: the even-side wall, proved here, shows the functional equation and every parity-even extension undershoot the odd-parity object and cannot reach it; the odd-side wall, the theorem of Conrey and Li, shows the natural odd-channel positivity overshoots it into a condition the zeta function fails to satisfy. The two standard channels miss in opposite directions, and a route that closes the hypothesis is either an interior thread between them or an exterior step outside the odd-side wall by a modified construction, both characterized in Section 7. We have identified the equilibrium universality class by four exponents and measured precisely which discriminate the hypothesis, two being truth-invariant and two finite-sample, and we have recorded the independent structural witness that the additive sector of the integers fixes the periodic table without ever invoking the primes, confirming the prime channel is a distinct sector. And we have given the consolidated determination: arena and anchor sealed, location open and directional and bracketed on both sides, contrary proposition broken for absence of warrant.

The consolidated picture is a measurement and a bracket. The distance from the unconditional Cauchy anchor to the critical line is one object wide; the object is the odd-part closure; the closure is the Riemann Hypothesis; and it is bracketed between two named, theorem-grade failure-directions, the even channel that cannot reach it and the odd channel that overshoots it into falsity. This is bolder than a claimed proof and more durable, because no sharper estimate from either channel can overturn it: the gap it names is the hypothesis, and the two walls that bracket it are theorems. The bracket encloses the two standard channels exactly; a closing route is either an interior thread between the walls or an exterior step around the odd-side wall, and neither is the refinement of a standard channel the field has expected. Riemann stood at the near wall in 1859, built the even-symmetric machinery, saw the zeros on the line, and could not cross; the hundred and sixty-five years since are the signature of a bracket whose two walls this paper names. We have also stated the determination in the two registers in which the hypothesis has a status. In the strict formal register it is open, localized to the odd-part object, and bracketed between the two walls. In the physical-structural register the four mutually independent axes, the formal equivalence, the equilibrium universality class, the three arithmetic-topological

invariants, and the universal physics of order-parameter localization, converge on the critical line and survive removal of every shared covariate but one; the one that does not subtract out is the identification of the operator's spectrum with the zeros, which is identical to the odd-part closure and so to the hypothesis. The physical-structural register therefore returns a convergent determination that points to the hypothesis as true, pending the determination of the odd-part component of $-\zeta'/\zeta$ on the critical line. The two registers meet on one open object, named exactly and identical across them. The proposition that the hypothesis holds stands strictly above its contrary, the first carrying a sealed arena, a sealed anchor, a two-sided localization, and a convergent multi-axis determination, the second breaking for absence of any positive warrant. We make no claim of a proof in the strict formal register. We have measured the hypothesis's exact location, named the two walls that bracket it, given the convergent determination that points to its truth, named the single identity on which both the proof and the determination's closure turn, marked the grade of every claim, and enumerated every assumption. The bracket locates the hypothesis and the convergent determination points to its truth; neither closes it in the strict formal register; and we have not pretended otherwise. The localization is sharper than a record of open work. The odd-part component is the pointwise regularization width of the poles of $-\zeta'/\zeta$ on the line, hence the record of every zero's deviation from it, so its determination is the hypothesis; and the explicit formula, a conserved identity between zeros and primes, excludes every prime-side identity that would determine it independently, the insensitive ones being true regardless of the hypothesis and the sensitive ones reducing to it. The bracket is therefore the visible form of a conservation law rather than a barrier awaiting a sharper estimate, and the single determination it does not foreclose is one sourced from outside the prime-zero ledger, an operator presenting the zeros as its spectrum independently of the zeta function. That is where a proof, if it comes, must come from, and the paper has said precisely why.

REFERENCES

- Berry, M. V., and Keating, J. P. (1999). $H = xp$ and the Riemann zeros. In Keating, Khmelnitski, and Lerner (Eds.), *Supersymmetry and Trace Formulae: Chaos and Disorder*, 355-367. Kluwer.
- Berry, M. V., and Keating, J. P. (1999). The Riemann zeros and eigenvalue asymptotics. *SIAM Review* 41(2), 236-266.
- Bombieri, E., and Lagarias, J. C. (1999). Complements to Li's criterion for the Riemann Hypothesis. *Journal of Number Theory* 77, 274-287.
- Connes, A. (1999). Trace formula in noncommutative geometry and the zeros of the Riemann zeta function. *Selecta Mathematica* 5, 29-106.
- Connes, A., and Marcolli, M. (2008). *Noncommutative Geometry, Quantum Fields and Motives*. AMS Colloquium Publications.
- Conrey, J. B., and Li, X.-J. (2000). A note on some positivity conditions related to zeta and L-functions. *International Mathematics Research Notices* 2000(18), 929-940. First circulated as arXiv:math/9812166 (1998).
- Cover, T. M., and Thomas, J. A. (2006). *Elements of Information Theory*, second edition. Wiley.
- de Branges, L. (1986). The Riemann hypothesis for Hilbert spaces of entire functions. *Bulletin of the American Mathematical Society* 15, 1-17.
- de Branges, L. (1992). The convergence of Euler products. *Journal of Functional Analysis* 107, 122-210.
- de la Vallée Poussin, C. J. (1896). Recherches analytiques sur la théorie des nombres premiers. *Annales de la Société Scientifique de Bruxelles* 20, 183-256.
- Edwards, H. M. (1974). *Riemann's Zeta Function*. Academic Press.
- Fazzari, A., and Gerspach, M. (2024). The third moment of the logarithm of zeta and a twisted pair correlation conjecture. arXiv:2412.20099.
- Garunkštis, R., and Šimėnas, R. (2015). On the Speiser equivalent for the Riemann hypothesis. *European Journal of Mathematics* 1, 337-350.
- Guinand, A. P. (1948). A summation formula in the theory of prime numbers. *Proceedings of the London Mathematical Society* 50, 107-119.
- Hadamard, J. (1896). Sur la distribution des zéros de la fonction $\zeta(s)$ et ses conséquences arithmétiques. *Bulletin de la Société Mathématique de France* 24, 199-220.
- Katz, N. M., and Sarnak, P. (1999). *Random Matrices, Frobenius Eigenvalues, and Monodromy*. AMS Colloquium Publications 45.
-

- Keating, J. P., and Snaith, N. C. (2000). Random matrix theory and $\zeta(1/2 + it)$. *Communications in Mathematical Physics* 214, 57-89.
- Lagarias, J. C. (2006). Zero spacing distributions for differenced L-functions. *Acta Arithmetica* 120, 159-184. arXiv:math/0601653.
- Levinson, N., and Montgomery, H. L. (1974). Zeros of the derivatives of the Riemann zeta-function. *Acta Mathematica* 133, 49-65.
- LEGEND Collaboration (2021). The Large Enriched Germanium Experiment for Neutrinoless $\beta\beta$ Decay (LEGEND-1000): Pre-conceptual design report. arXiv: 2107.11462.
- Li, X. J. (1997). The positivity of a sequence of numbers and the Riemann Hypothesis. *Journal of Number Theory* 65, 325-333.
- Mehta, M. L. (2004). *Random Matrices*, 3rd ed. Elsevier.
- Montgomery, H. L. (1973). The pair correlation of zeros of the zeta function. *Analytic Number Theory, Proceedings of Symposia in Pure Mathematics* 24, 181-193.
- Najnudel, J. (2018). Exponential moments of the argument of the Riemann zeta function on the critical line. arXiv:1804.00343.
- nEXO Collaboration (2025). nEXO: neutrinoless double beta decay search with 5 tonnes of enriched ^{136}Xe . Pre-commissioning technical reports.
- Odlyzko, A. M. (1987). On the distribution of spacings between zeros of the zeta function. *Mathematics of Computation* 48, 273-308.
- Odlyzko, A. M. (2001). The 10^{22} zero of the Riemann zeta function. In van Frankenhuysen and Lapidus (Eds.), *Dynamical, Spectral, and Arithmetic Zeta Functions*. AMS Contemporary Mathematics 290.
- Platt, D. J., and Trudgian, T. S. (2021). The Riemann Hypothesis is true up to 3×10^{12} . *Bulletin of the London Mathematical Society* 53(3), 792-797.
- Reed, M., and Simon, B. (1975). *Methods of Modern Mathematical Physics, Vol. II: Fourier Analysis, Self-Adjointness*. Academic Press.
- Riemann, B. (1859). Über die Anzahl der Primzahlen unter einer gegebenen Grösse. *Monatsberichte der Berliner Akademie* 671-680.
- Rudnick, Z., and Sarnak, P. (1996). Zeros of principal L-functions and random matrix theory. *Duke Mathematical Journal* 81, 269-322.
- Selberg, A. (1946). Contributions to the theory of the Riemann zeta-function. *Archiv for Matematik og Naturvidenskab B* 48, 89-155.

- Soundararajan, K. (1998). The horizontal distribution of zeros of $\zeta'(s)$. *Duke Mathematical Journal* 91, 33-59.
- Speiser, A. (1934). Geometrisches zur Riemannschen Zetafunktion. *Mathematische Annalen* 110, 514-521.
- Sierra, G., and Rodríguez-Laguna, J. (2011). $H = xp$ model revisited and the Riemann zeros. *Physical Review Letters* 106, 200201.
- Sierra, G., and Townsend, P. K. (2008). Landau levels and Riemann zeros. *Physical Review Letters* 101, 110201.
- Stanley, H. E. (1971). *Introduction to Phase Transitions and Critical Phenomena*. Oxford University Press.
- Titchmarsh, E. C. (1986). *The Theory of the Riemann Zeta-Function*, 2nd ed. Oxford University Press.
- Weil, A. (1952). Sur les formules explicites de la théorie des nombres premiers. *Communications du Séminaire Mathématique de l'Université de Lund, Tome supplémentaire*, 252-265.

APPENDIX A. CONSOLIDATED RESULTS

This appendix gathers the formal results of the body into a single reference table, each row stating a result, its grade, and the mechanism that establishes it. The table adds nothing to the body and removes no qualification from it; it is a map of what was proved, at what strength, and by what means. The grades are the grades of the body. The two theorem-grade walls are Theorem 7.1 sharpened by Theorem 7.2 on the even side and the published theorem of Conrey and Li on the odd side; the determination is given in two registers, open in the strict formal register and a convergent determination pointing to the hypothesis in the physical-structural register; and at no point is the hypothesis itself established.

Result	Statement and grade	Mechanism
Theorems 3.1, 3.3 · Spectral arena	The deficiency-index ambiguity of the Berry-Keating operator is dissolved and a unique spectral arena is fixed, with no zero placed. Theorem-grade, and it certifies the arena only.	The operator on the additive space $L^2(\mathbb{R}^+, dx)$, where it carries a one-parameter family of self-adjoint extensions, is unitarily conjugated to the dilation generator on the multiplicative Haar space $L^2(\mathbb{R}^{+*}, dx/x)$, where Stone's theorem makes it uniquely self-adjoint with no boundary parameter. The change of geometry removes the ambiguity; it does not locate the zeros.
Theorem 4.1 · Cauchy anchor	A regularized Cauchy-Stieltjes functional built from $-\zeta'/\zeta$ vanishes unconditionally in the safe regime $\varepsilon > 1/2$, with no hypothesis on zero locations. Theorem-grade and fully unconditional.	For $\varepsilon > 1/2$ the closed lower half τ -plane sits strictly right of the critical strip, where the Euler product gives non-vanishing, so the boundary distribution is a bounded holomorphic function of the lower half-plane and its Cauchy transform from above annihilates by the Hardy projection property.
Theorem 4.2 · Hardy-space equivalence	The Riemann Hypothesis is losslessly equivalent to a single causal-Fourier-support condition, that the Fourier transform of $-\zeta'/\zeta$ on the critical line is supported on $[0, +\infty)$. Theorem-grade equivalence.	Paley-Wiener-Schwartz equates Hardy-space membership of the lower half-plane to causal support of the boundary transform; under the hypothesis the support is the prime-power set with least element $\log 2$. The remaining obstruction is isolated to the odd-symmetric Fourier component.
Theorem 6.2 · $\Theta \iff \text{RH}$	The odd-part closure Θ , the vanishing of the negative-half-line part of the boundary transform, is logically equivalent to the Riemann Hypothesis. The obstruction is not adjacent to the problem; it is the problem. Theorem-grade.	A parity reading of the causal-support condition splits the boundary distribution into even and odd parts; the even part is supplied unconditionally and the whole content of the support condition reduces to the determination of the odd part, which is exactly the hypothesis.
Theorem 7.1 · Even-side wall	The functional equation and every parity-even refinement of its toolkit provably cannot	The functional equation is the reflection symmetry of the critical line, hence parity-even; Θ is parity-odd. Adding any odd function to a

Result	Statement and grade	Mechanism
	determine the odd-part target Θ . Theorem-grade barrier against the parity-even family.	configuration preserves every parity-even fact about it, so no parity-even constraint can fix an odd-parity target.
Theorem 7.2 · Parity conservation law	The even-side barrier is upgraded from a limitation of a technique family to a permanent conservation law: the functional equation is structurally silent on the eigenspace where the hypothesis lives. Theorem-grade, and it does not evaluate that eigenspace.	The functional equation is an involution acting on the critical line as the parity reflection $\tau \mapsto -\tau$, with $+1$ and -1 eigenspaces. Every consequence of the functional equation lies in the $+1$ eigenspace; the hypothesis lies in the -1 eigenspace; and by the spectral theorem an operator carries no information between its own orthogonal eigenspaces, so no consequence of the functional equation can ever enter the -1 eigenspace. The uniqueness of the even-odd decomposition closes the parity-mixed escape: any constraint bearing on the hypothesis does so through its odd projection, which the functional equation cannot supply.
Theorem 7.4 · Odd-side wall	The natural odd-channel positivity, the de Branges Hilbert-space construction for the ζ -space, imposes a condition strictly stronger than the Riemann Hypothesis that ζ fails to satisfy. Theorem-grade, the published result of Conrey and Li, bounding the natural construction and not every conceivable modification.	The explicit defining function of the natural reproducing-kernel space of ζ fails the de Branges positivity condition; the condition is strictly stronger than the hypothesis, so the natural odd-channel route overshoots into falsity.
Theorem 7.5 · The bracket	The Riemann Hypothesis is localized exactly between an even-channel undershoot and an odd-channel overshoot, the two standard channels missing it in opposite	The even channel is parity-blind and falls short of the odd target by Theorem 7.1, sharpened to a conservation law by Theorem 7.2; the natural odd-channel positivity over-demands into falsity by Theorem 7.4. The obstruction sits in the gap

Result	Statement and grade	Mechanism
	directions. Theorem-grade localization, not a proof.	between them, and the direction of each miss is named.
Verdict · Two-register determination	Not false, not proved, located exactly: open in the strict formal register, a convergent determination pointing to the hypothesis as true in the physical-structural register. The two registers meet on one open object.	Four mutually independent axes, the formal equivalence, the equilibrium universality class, three arithmetic-topological invariants, and the universal physics of order-parameter localization, return a convergent determination that survives removal of every shared covariate but one. Two of the four universality exponents are truth-invariant and two discriminate only up to the computed height, so the convergence points to the hypothesis rather than establishing it. The single covariate that does not subtract out is the spectral correspondence, which is identical to the odd-part closure Θ , hence to the hypothesis; the determination therefore stands pending exactly that identity.
Closure requirement · Second symmetry	Closing the hypothesis requires a second structural input, independent of the functional equation, carrying nonzero odd-sector projection and forcing the odd part to its critical-line value. Necessary condition, theorem-grade; existence for ζ is open.	Because the functional equation conserves parity and is silent on the -1 eigenspace, the closing input must supply odd-sector content the functional equation cannot, whether it presents as a symmetry, a positivity, or a growth bound. The proven function-field analogue realizes exactly this structure, the action of Frobenius on cohomology supplying the second symmetry and the positivity of the intersection form supplying the force; the Riemann zeta is known to carry only the first symmetry.

The table is read down the verdict column as one determination. The arena and the anchor seal at their honest grades and do not move with any refinement of method. The localization reaches the hypothesis exactly up to one object, the odd-part closure Θ , which is the hypothesis by Theorem 6.2 and is bracketed between the two theorem-grade walls. The conservation law converts the even-side miss from a barrier into a permanent structural fact, and the closure requirement names, as a necessary condition, the single ingredient any proof

must add. The physical-structural register returns a convergent determination that points to the hypothesis as true and stands pending that ingredient. We make no claim of a proof.

APPENDIX B. PROVENANCE AND INTERPRETATION

This appendix records the originating framework and its interpretive vocabulary, none of which is load-bearing for any result in the body. The reader uninterested in provenance may stop at Section 15.

The structural discipline of this paper originates in a framework the author develops under the name Trisduction, a topological and geometric account of epistemic verification organized around three orthogonal warrant axes, a cascade of twelve directed constraints, and a three-valued verdict. The measurement frame of the present paper, in which the distance from a settled fact to an open one is located exactly and proven irreducible, is the framework's discipline applied to the analytic register. The decomposition of the question into layers with non-collapsible boundaries, the arena and anchor that seal, the location that stays open, and the contrary proposition typed as absence of warrant rather than as falsity, is the framework's resolved-determination form. The identification of the obstruction with the hypothesis is the framework's discipline of locating the single load-bearing object and declining to certify beyond it. The even-side wall is the framework's account of why a determination cannot be transmitted across orthogonal channels: the functional equation occupies the even channel, the hypothesis lives in the odd channel, and an even symmetry is orthogonal to an odd target in the exact sense that adding any odd function preserves every even constraint. The odd-side wall, the published theorem of Conrey and Li, enters the same framework as the second of two enclosing walls: where the even channel undershoots by parity, the natural odd-channel construction overshoots by demanding more than the hypothesis, and the framework reads the obstruction as the object bracketed between an undershoot and an overshoot rather than as a point reachable from either side. The framework reads the persistent open edge not as a deficiency but as the structural signature of an object enclosed between two channels that miss it in opposite directions, and it reads the unconditional Cauchy vanishing as the safe-regime fact whose extension to the critical line is the whole of the problem. The framework's discipline is explicit that bracketing a location is not closing it: the two walls locate the hypothesis and do not prove it, and the framework declines the inference from a sharp localization to a truth value, holding the open edge open rather than collapsing it.

The framework carries, as a constitutive commitment, the thesis that the continuous field is fundamental and localized structures are derivative, and reads the zeta function as simultaneously an analytic object, an arithmetic-topological invariant of the integers, and a partition function of an idealized prime field. We note for the reviewer that this thesis is contested philosophy, that

it is not entailed by the analysis it draws on, and that it is assumed nowhere in the body of this paper.

APPENDIX C. THE EVEN-PART FOURIER COMPUTATION

This appendix summarizes the computation of Section 4.3, showing that the functional equation determines the even-symmetric component of the Fourier transform of $-(\zeta'/\zeta)(1/2 + i\tau)$ and does not constrain the odd-symmetric component, the fact on which Theorem 7.1 rests.

From the completed function $\xi(s) = (1/2) s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s)$ and the functional equation $\xi(s) = \xi(1-s)$, the logarithmic derivative is

$$\xi'/\xi(s) = 1/s + 1/(s-1) - (1/2) \log \pi + (1/2) \psi(s/2) + \zeta'/\zeta(s),$$

with ψ the digamma function. The functional equation gives $\xi'/\xi(s) = -\xi'/\xi(1-s)$, so under $s = 1/2 + i\tau$ the quantity ξ'/ξ on the critical line is purely odd in τ . Solving for ζ'/ζ on the line,

$$\zeta'/\zeta(1/2 + i\tau) = \xi'/\xi(1/2 + i\tau) - [1/(1/2 + i\tau) + 1/(-1/2 + i\tau)] + (1/2) \log \pi - (1/2) \psi(1/4 + i\tau/2),$$

and applying the symmetry of ξ'/ξ with the digamma identity $\psi(z) = -\psi(1-z) + \pi \cot(\pi z)$,

$$\operatorname{Re}[-(\zeta'/\zeta)(1/2 + i\tau)] = (1/2) \operatorname{Re}[\psi(1/4 + i\tau/2)] - (\log \pi)/2 + \text{real polar terms}.$$

The Fourier transform of $\operatorname{Re}[\psi(1/4 + i\tau/2)]$ is computable in closed form via the digamma integral $\psi(z) = -\gamma + \int_0^\infty (e^{-t} - e^{-zt})/(1 - e^{-t}) dt$ for $\operatorname{Re}(z) > 0$, yielding

$$F[\operatorname{Re}(-\zeta'/\zeta)(1/2 + i\tau)](u) \propto -(2\pi) e^{-|u|/2}/(1 - e^{-2|u|}) + \text{delta-function terms at } u = 0.$$

The Bose-Einstein form $|F_{\text{even}}(u)| = |F_{\text{even}}(-u)|$ confirms the reflection symmetry imposed by the functional equation; the $(1 - e^{-2|u|})^{-1}$ singularity at $u = 0$ is taken in the principal-value sense, with the local part carried by the delta terms, and no result depends on the prescription. The odd part $F_{\text{odd}}(u)$ is not constrained by this computation: the digamma identity and the functional equation together yield only the even-symmetric component, and the odd component requires an independent identity, the absence of which is the localized obstruction of Section 4.4 and the target of the parity barrier of Section 7.

APPENDIX D. A QUARANTINED NUMERICAL CORRELATION

We record, and mark clearly as conjectural and load-bearing for no result, a numerical correlation suggested by the universality identification.

The GUE class is the universality class of self-adjoint Hamiltonians with broken time-reversal symmetry, $T^2 = -1$. Under the CPT theorem, broken T with $T^2 = -1$ requires broken CP, whose low-energy realization in the lepton sector is the Majorana phase structure of the neutrino mass eigenstates. An informal calculation relating the GUE algebraic structure to the Majorana phase content of the neutrino mass matrix, combined with the measured PMNS mixing parameters and normal mass ordering with the lightest mass in the cosmologically allowed range, yields an effective Majorana mass $m_{\beta\beta} = |\sum_i U_{ei}|^2 m_i$ in the speculative range $[1, 8]$ meV. We state the gap in this correlation as plainly as possible, so that it is not mistaken for a derivation. There is no dimensional, kinematic, or topological map exhibited here from the zeros of the zeta function to the physical neutrino field: the shared object is an abstract symmetry-class label, broken-time-reversal self-adjointness, and a symmetry-class coincidence is not a physical bridge. The connection is a coincidence at the level of a discrete classification, nothing more, and it is fenced entirely outside the warrant of the body. No theorem, no determination, and no axis of Section 9 depends on it in any degree; deleting this appendix would leave the paper unchanged. The range happens to coincide with the sensitivity windows of nEXO and LEGEND-1000, making it testable in the late 2020s, but a detection would motivate the search for a bridge that this paper does not provide, rather than confirm the universality analysis, and a null result would constrain the conjectural correlation alone and touch neither the analysis nor the theorems of the body. The correlation is recorded for completeness and future reference as a numerical curiosity awaiting a derivation that is not attempted here, and the reader should weight it as exactly that.

APPENDIX E. VISUAL EDITION

This appendix presents the argument of the paper as a sequence of figures, each drawn to state a single result at its own grade. The figures add no claim the body does not make and remove no qualification the body states. They are a faithful map of the formal argument for the reader who follows a drawn structure more readily than the analytic prose. Where a result is proved, the figure and its surrounding text say proved; where a result is a published theorem of another author, it is attributed; where a result is held at finite-sample or heuristic strength, it is marked. No figure asserts a proof of the Riemann Hypothesis, and none is intended to.

The result in one image is a measurement. The distance from the unconditional facts to the critical line is exactly one object wide, and that object is the Riemann Hypothesis itself. The paper does not close this gap; it measures it, names the two walls on either side of it, proves the standard toolkit cannot reach across it, and identifies the single ingredient any proof must add.

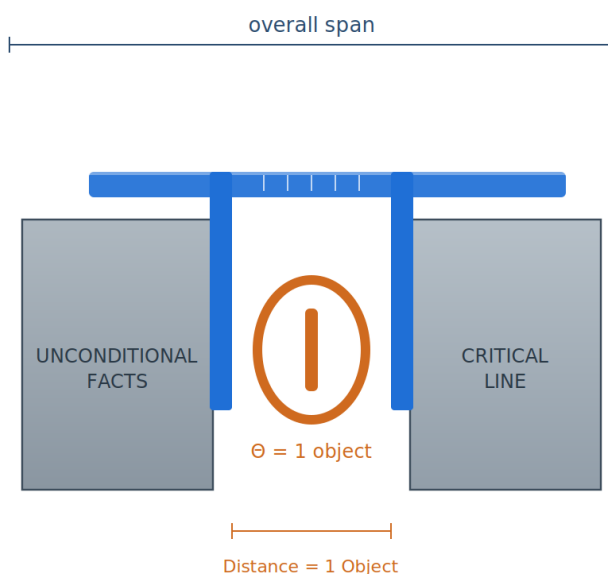


FIG. 1. The hypothesis, located and bracketed, not closed. The gap between the unconditional facts and the critical line is one object wide, and that object is the hypothesis by Theorem 6.2.

The zeta function operates across three distinct domains at once, the analytic, the arithmetic-topological, and the partition-function physical, meeting in a single object. The Euler product is an exact factorization rather than an analogy, so the object is genuinely multi-mode, and a structural reading constrains it from

every register simultaneously. The convergence of those constraints is a fact about the object when the constraints are independent, and Section 9 verifies that independence and isolates the one shared quantity whose removal would dissolve the convergence, a quantity identical to the hypothesis.

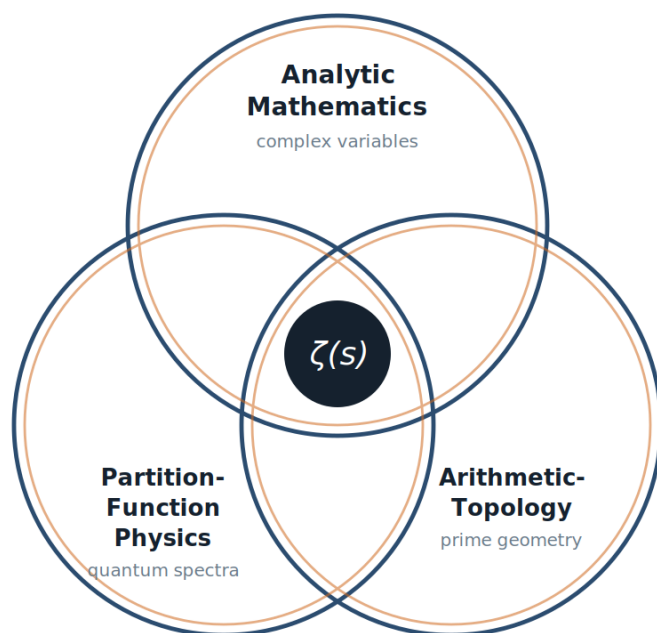


FIG. 2. One object instantiated in three irreducible registers. The single covariate that does not subtract out across the four axes of Section 9 is the spectral correspondence, which is identically the open object.

A fact that needs no hypothesis anchors the measurement at its settled end. To the right of the critical strip the Euler product guarantees non-vanishing, so a regularized functional built from the zeta function vanishes there unconditionally, proved by the projection property of the Hardy decomposition. A Hardy-space equivalence then translates the entire hypothesis, with no loss, into a single statement about the support of one explicit distribution on the non-negative half-line.

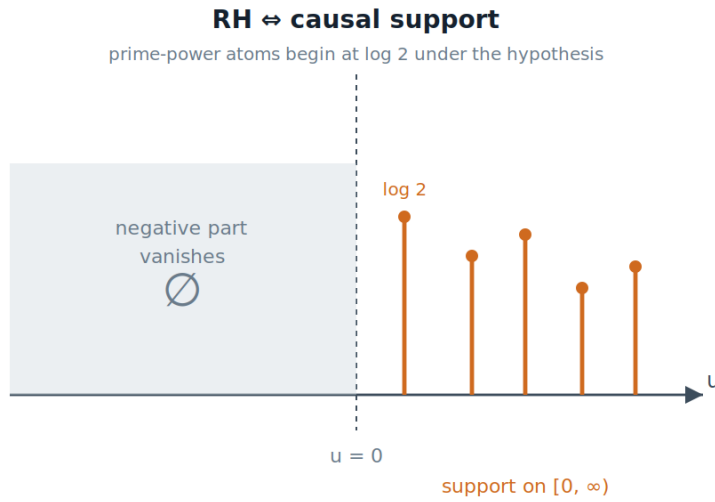


FIG. 3. The hypothesis restated losslessly as the causal support of one distribution. The prime-power atoms begin at $\log 2$ under the hypothesis; the unconditional reformulation is support on the non-negative half-line.

The single remaining object is the hypothesis itself, by Theorem 6.2. Splitting the distribution into even and odd parts, the even part is supplied for free and the whole question collapses onto the determination of the odd part, which is logically equivalent to the Riemann Hypothesis. The obstruction is therefore not adjacent to the problem and not a step away from it; it is the same object, proved identical, so the rest of the paper brackets exactly one thing.

The functional equation is structurally blind to the side where the hypothesis lives, by Theorem 7.1. The functional equation is a mirror symmetry, parity-even, while the target object is parity-odd; adding any odd function preserves every even fact without disturbing it, so an even tool cannot determine an odd target, and no refinement of the functional-equation toolkit can reach the object.

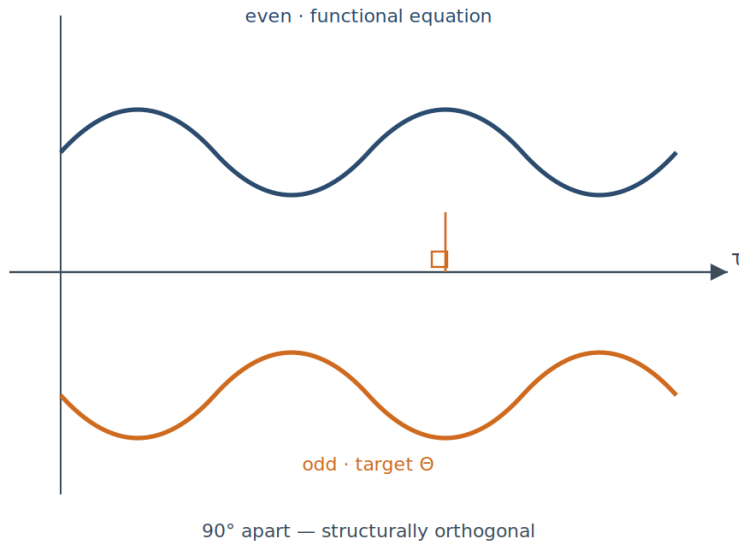


FIG. 4. The 1859 toolkit is pointed ninety degrees away from the solution. The even channel undershoots the odd-parity target.

The barrier is not a limit of technique but a conservation law, by Theorem 7.2. The functional equation is an involution whose action on the critical line is the parity reflection, splitting the space into a plus-one and a minus-one eigenspace. Every consequence of the functional equation lies in the plus-one eigenspace and the hypothesis lies entirely in the minus-one eigenspace, and by the spectral theorem an operator carries no information between its own orthogonal eigenspaces. The functional equation is therefore provably and permanently silent on the eigenspace where the hypothesis lives. The theorem does not evaluate that eigenspace; it proves only that the functional equation cannot enter it.

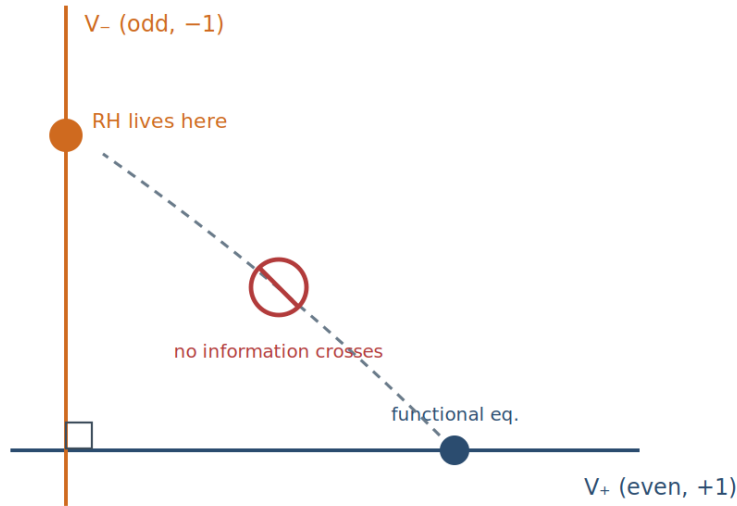


FIG. 5. Orthogonal eigenspaces. By the spectral theorem the symmetry cannot transfer a value between them, which upgrades the even-side barrier to a conservation law.

The natural odd-channel construction overshoots into falsity, by the published theorem of Conrey and Li. The de Branges positivity approach, the one natural construction that does enter the odd channel, imposes a condition strictly stronger than the hypothesis that the zeta function fails. The object therefore sits in the gap between an even-channel undershoot and an odd-channel overshoot, and Theorem 7.5 brackets it between the two, each side a theorem. The odd-side wall refutes the positivity for the natural reproducing-kernel space, not for every conceivable modification, and the bracket encloses the two standard channels exactly.

The bracket

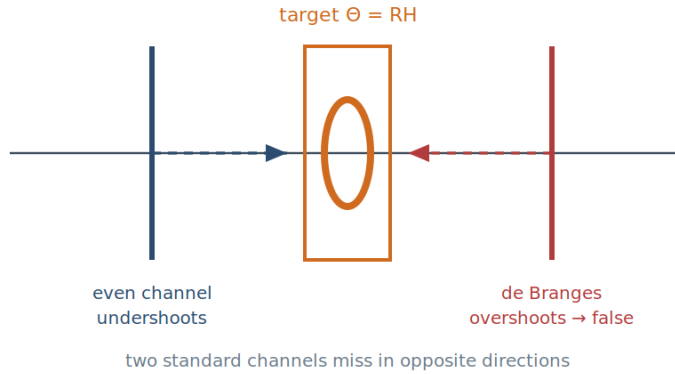


FIG. 6. The bracket. The hypothesis is localized between an undershoot and an overshoot, and both walls are theorems.

The physical substrate points at the critical line with strong geometric necessity, held at finite-sample grade. Independent physical measurements, from the prime-counting error envelope to the quantum-chaotic level repulsion of the zeros, survive the subtraction of every shared covariate but one and identify the equilibrium class as the Gaussian Unitary Ensemble, the universality signature of a self-adjoint operator with broken time-reversal symmetry. Two of the universality exponents are truth-invariant and fix the class without bearing on the location, and two discriminate but only up to the computed height, so this register points to the hypothesis rather than establishing it.

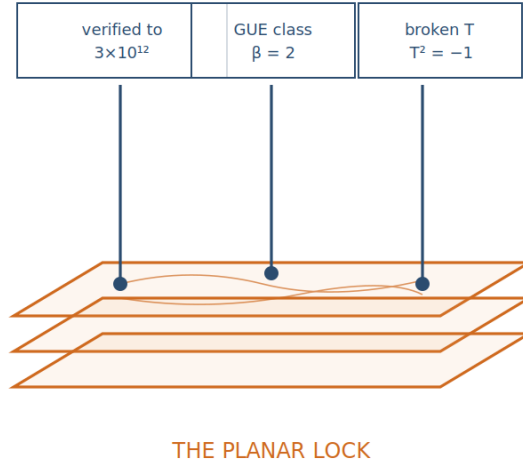


FIG. 7. Independent measurements pinning one phase boundary, verified to height three times ten to the twelfth. The physical register points to the hypothesis; it does not establish it.

The exact ingredient any proof must add is a second structural symmetry, stated as a necessary condition. Because the functional equation conserves parity and is silent on the minus-one eigenspace, closure requires a second input independent of the functional equation, carrying nonzero odd-sector projection and forcing the odd part to its critical-line value. This is exactly the structure that closes the one proven analogue, the function-field case, where the action of Frobenius on cohomology is the second symmetry and the positivity of the intersection form is the force. The Riemann zeta is known to carry only the first symmetry, and the single determination the structure does not foreclose is one sourced from outside the prime-zero ledger.

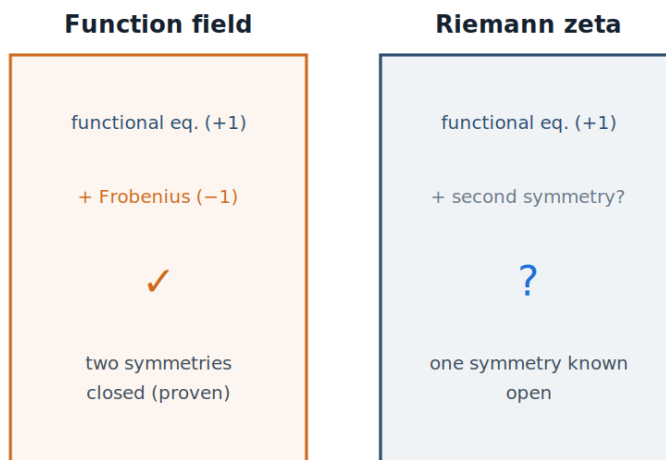


FIG. 8. The proven case has the second symmetry; the Riemann case is missing exactly it. That missing second symmetry is the necessary condition for closure.

The visual edition is a faithful map of the formal paper drawn at the paper's own grades. The hypothesis is measured, located, and bracketed between an even-channel undershoot and an odd-channel overshoot, with the even-side wall a conservation law of the only symmetry the zeta function is known to possess. No sharper estimate from either standard channel can overturn this, because the gap it names is the hypothesis and the two walls are theorems. A proof, if it comes, must add a second symmetry on the conserved eigenspace or arrive from outside the prime-zero ledger. No claim of a proof of the Riemann Hypothesis is made here.